

EcoPin A.I.: Crowdsourced Geospatial Platform for Transparent Environmental Reporting and Rapid Institutional Detection for the Pasig City Solid Waste Management Office

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ABSTRACT

CHAPTER 1

Introduction

The Problem and its Background

Urban environmental issues such as improper waste disposal and pollution continue to challenge local government units (LGUs). Although citizens can report these concerns through hotlines, social media, emails, and other channels, environmental information remains scattered across multiple platforms, a fragmentation compounded by weak enforcement and inconsistent monitoring practices documented across Philippine LGUs (Coracero et al., 2021). This makes it difficult for authorities to monitor problem locations, identify recurring issues, and respond promptly, often allowing problems to persist or recur before institutional action is taken.

Geographic Information Systems (GIS) and crowdsourced reporting have shown significant potential in improving environmental monitoring. The MASDAN platform demonstrated how GIS and citizen participation can facilitate real-time environmental reporting and strengthen communication between communities and government agencies (Narciso et al., 2021). However, existing systems like MASDAN stop at collection and visualization, offering limited support for validating submitted evidence, clustering related reports, or forecasting emerging hotspots, leaving institutions with visibility but no built-in mechanism for prioritization or trust in the data they receive. Recent studies also indicate that urban environmental monitoring in the Philippines remains an area requiring

further development despite the increasing adoption of geospatial technologies (Salvacion, 2025).

To close this integration gap, this study proposes EcoPin A.I., a crowdsourced environmental reporting platform that combines GIS, AI, and community participation. Residents can submit geotagged reports with images and descriptions; AI validates uploaded images for authenticity, and geospatial analytics cluster and forecast environmental hotspots to support faster, evidence-based institutional response. Initially implemented around Pamantasan ng Lungsod ng Pasig (PLP), the system aims to improve reporting efficiency, strengthen collaboration between citizens and LGUs, and support more proactive environmental management.

Review of Related Literature

Understanding how EcoPin A.I. fills a real gap requires looking at three converging problems, the institutional and behavioral barriers that have stalled solid waste management in the Philippines, the geospatial and citizen-science scholarship explaining how participatory reporting can be made to work, and the emerging use of artificial intelligence to add analytical value that current civic platforms still lack.

Solid Waste Management Gaps and Institutional Enforcement in the Philippines

The persistence of solid waste mismanagement in urban areas of the Philippines has been traced to a combination of expanding waste volumes and uneven enforcement of policy. According to a nationwide review of the state of solid waste management, weak implementation of the Ecological Solid Waste Management Act (RA 9003), a scarcity of sanitary landfills, and widespread improper disposal are recurring problems across both urban and rural local government units (Coracero et al., 2021).

Their review further established that technical and infrastructural fixes are insufficient on their own. Active citizen involvement was identified as a necessary complement to government intervention. This conclusion frames the central premise of the present study that a platform's value depends as much on how it mobilizes community participation as on the sophistication of its backend systems.

Participatory GIS Platforms for Community-Based Environmental Reporting

Building on the recognized need for citizen involvement, several local platforms have attempted to operationalize participatory reporting through geographic information systems. According to the developers of the MASDAN application, the system paired GIS mapping with a citizen-reporting interface that let residents flag environmental concerns affecting Manila Bay and allowed local authorities to visualize the resulting reports (Narciso et al., 2021).

Their work demonstrated that a two-way information channel between citizens and institutions is technically achievable in a Philippine setting. The system's functions, however, were confined to data collection and visualization. Predictive analysis, automated validation, and workload-based prioritization were not part of its design, a limitation that subsequent themes in this review revisit.

Geospatial Technologies in Environmental Planning and Monitoring

Beyond single-platform examples, the broader adoption of geospatial technology across Philippine sectors offers context for where GIS-based environmental monitoring currently stands. According to a study tracing GIS adoption across the country, there has been a marked increase in GIS use across agricultural, disaster-risk, and environmental research in the country, while also flagging urban environmental monitoring as comparatively underdeveloped relative to these other domains (Salvacion, 2025).

The same study stressed that geospatial transparency, meaning open and consolidated access to spatial information, supports more informed planning and resource allocation by decision-makers. Together with the earlier-cited findings on MASDAN (Narciso et al., 2021), this body of work confirms that the technical groundwork for geospatial citizen reporting already exists in the Philippines, but that its application to everyday urban waste and environmental concerns remains a comparatively open area.

Citizen Science and Crowdsourced Monitoring in Resource-Limited Urban Settings

A parallel and growing literature outside the Philippines speaks to how citizen-contributed data performs under the resource constraints typical of developing urban contexts. According to a citizen science flood-monitoring program implemented within informal urban settlements in Indonesia and Fiji, resident-submitted observations filled monitoring gaps within informal urban settlements in Indonesia and Fiji and found that resident-submitted observations filled monitoring gaps left by institutions lacking the staff or sensor infrastructure to cover these areas continuously (Wolff et al., 2021).

Their study underscored that community members are capable of producing usable environmental data through a simple reporting mechanism, provided the platform accounts for local infrastructure limitations such as inconsistent internet access. This finding lends empirical support to the assumption underlying EcoPin A.I., namely that ordinary residents, not only trained personnel, can serve as a reliable first layer of environmental detection.

Speed and Responsiveness of Digital Citizen Reporting Platforms

Where the previous theme established that citizens can supply usable data, a related question concerns whether such data translates into faster institutional response. According to a comparison of resolution times for collaborative, citizen-submitted reports against conventional non-collaborative requests on digital civic platforms and found that the collaborative reports were resolved markedly faster (Schiff, 2025).

This timing advantage speaks directly to the second specific problem this study investigates, namely the lead time that can be gained when problems are first detected through citizen reports rather than through scheduled patrols. Read alongside the earlier-cited work on geospatial transparency (Salvacion, 2025) the evidence suggests that pairing geospatial visibility with a collaborative reporting structure does not merely make problems easier to see. It also appears to shorten the interval between a problem's emergence and an institution's awareness of it.

Synthesis

The first five themes converge on a single proposition, weak enforcement and fragmented data persist in Philippine solid waste management (Coracero et al., 2021) not for lack of citizen willingness but for lack of a unifying technical mechanism.

Participatory GIS systems such as MASDAN have shown that residents will use a mapping interface to report concerns (Narciso et al., 2021), and the wider geospatial literature confirms that the infrastructure for such systems is maturing nationally even as urban applications lag (Salvacion, 2025). Evidence from comparable urban contexts abroad reinforces that crowdsourced data is dependable enough to substitute for, or supplement, institutional monitoring in areas patrols cannot reach continuously (Wolff et al., 2021), and platform-level evidence indicates that collaborative reporting structures measurably shorten response times (Schiff, 2025). None of these five studies, however, addresses what happens once reports accumulate.

Spatial Clustering and Hotspot Analysis for Prioritization

Once a platform succeeds in collecting geotagged reports at scale, the next challenge is translating individual points into actionable priorities. According to a survey of statistically robust clustering techniques for mapping spatial hotspots, such methods, originally developed for public health and public safety applications, are now applied across environmental and urban domains to identify regions with a disproportionately high concentration of events (Xie et al., 2022).

The same survey cautioned that without statistical rigor in defining a cluster, agencies risk dispatching resources toward spurious or coincidental groupings rather than genuine problem areas. For EcoPin A.I., this means clustering has to do more than group nearby pins on a map, it has to filter out the noise of isolated, one-off reports so that the clusters an LGU acts on represent a real, recurring concentration of problems rather than a coincidence of timing.

Artificial Intelligence for Image Validation in Waste and Environmental Management

A second technical gap concerns the credibility of submitted reports themselves. According to a review of the expanding role of artificial intelligence in smart-city waste management, deep learning-based image classification has been used to detect illegal dumping sites, sort waste types, and filter low-quality submissions before they reach human reviewers (Fang et al., 2023).

Their review observed that pre-trained classification models can substantially reduce the manual screening burden placed on institutions with limited inspection

staff, though accuracy was noted to still vary with image quality and environmental conditions such as lighting or occlusion. For EcoPin A.I., this body of work justifies positioning AI-based image validation not as a novelty feature but as the mechanism that allows institutions to trust crowdsourced submissions enough to act on them without re-inspecting every pin in person.

Frontline and Informal Waste Workers as Untapped Reporting Capacity

While platform design addresses visibility and validation, the people closest to waste problems on a daily basis are rarely formal contributors to digital systems. According to a 2023 study by the Department of Labor and Employment's Institute for Labor Studies, which examined just-transition and circular-economy outcomes for waste workers across six Philippine LGU sites piloting zero-waste programs, drawing on interviews with local government officials, materials recovery facility (MRF) managers, and formal and informal waste workers, persistent decent-work deficits were found spanning employment stability, social protection, skills training, and occupational health and safety, along with a notable absence of social dialogue mechanisms that would allow workers to participate in waste planning decisions (Son, 2023). The study used the International Labour Organization's Just Transition Guidelines as its analytical framework.

This lack of institutional voice is corroborated by academic research on informal recycling-chain actors in urban Philippine settings, according to which hazardous working conditions such as elevated heavy-metal exposure from open burning and acid-bath practices, alongside precarious incomes and exploitation by

middlemen, are documented, with these workers' systemic role in the recycling chain routinely excluded from formal monitoring or reporting structures (Romallosa et al., 2024). Taken together, these findings suggest that frontline and informal waste workers already function as de facto environmental observers through the sheer routine of their daily rounds.

What they lack is not proximity to the problem, but a structured, geotagged channel through which those observations could be formally captured and systematized.

Transparency, Public Dashboards, and Institutional Accountability

The fourth specific problem this study investigates concerns the effect of public access to report status on community awareness and institutional accountability. Philippine local governance has shown movement toward addressing this directly, according to the Quezon City government's commitment under the Open Government Partnership (2025) to optimize its public-facing data dashboard, an underused or inconsistently updated dashboard reduces citizen trust in its functionality and limits its value as a tool for accountability, while a properly maintained, real-time tracker linked to citizen-submitted concerns was expected to strengthen both transparency and public engagement.

This commitment indicates that Philippine LGUs already recognize public status-tracking as a governance objective in its own right, separate from the technical question of data collection. EcoPin A.I.'s decision to let residents view whether a pin is resolved, in-progress, or unresolved therefore responds to a

documented institutional aspiration that existing systems, including MASDAN, have not yet operationalized at the level of individual report status.

ISO/IEC 25010 as a Framework for Evaluating Civic Technology Systems

Finally, building a platform that addresses visibility, speed, prioritization, and transparency raises a separate question of how its software quality should be evaluated once developed. According to the ISO/IEC 25010:2023 product quality model, the current edition of the international standard, eight characteristics are defined against which an information system can be measured (International Organization for Standardization, 2023).

According to a bibliometric review tracing research applying this standard across public administration, finance, education, and healthcare, publication activity has risen steadily, peaking in 2023, indicating that ISO/IEC 25010 has become a default reference point for evaluating government-facing digital systems specifically, not only commercial software (Rojas et al., 2025). For EcoPin A.I., adopting this framework offers a structured basis for assessing whether the finished system, beyond simply working, actually meets the functional suitability, usability, and reliability standards expected of a platform that both citizens and LGUs will depend on.

Synthesis and Research Gap

The second cluster of themes addresses what happens after citizen reports are collected, data must be clustered into actionable priorities (Xie et al., 2022), screened for credibility through automated image validation (Fang et al., 2023), opened to the frontline workers who already perform informal monitoring (Son, 2023), made visible to the public in a way that sustains institutional accountability (Open Government Partnership, 2025), and ultimately evaluated against an internationally recognized quality standard (Rojas et al., 2025; International Organization for Standardization, 2023).

Where the first five themes established that citizen-contributed, geospatially organized data is both feasible and faster than traditional patrols, this second cluster demonstrates that feasibility alone does not guarantee institutional usefulness. A platform that collects thousands of unclustered, unvalidated pins with no public status tracker offers little improvement over the fragmented hotline and social-media reports it intends to replace.

Across the ten themes, the reviewed literature affirms that the building blocks for a unified environmental reporting platform already exist as separate, validated components. Participatory GIS reporting is technically proven in the Philippines (Narciso et al., 2021), geospatial technology is maturing nationally even where urban applications lag (Salvacion, 2025).

Crowdsourced citizen data performs reliably even in resource-limited settings (Wolff et al., 2021), collaborative platforms measurably shorten institutional

response time (Schiff, 2025). Statistically grounded clustering can convert scattered reports into defensible priority areas (Xie et al., 2022). AI-based image classification can screen submissions before they reach institutional queues (Fang et al., 2023). Frontline waste workers represent an underused source of ground-level monitoring (Son, 2023). Philippine LGUs are actively pursuing public-facing accountability dashboards (Open Government Partnership, 2025) and ISO/IEC 25010 offers a recognized standard against which the resulting system can be measured (Rojas et al., 2025).

What is absent from the literature is a single platform that integrates all of these components for one coherent purpose. MASDAN demonstrates participatory GIS but stops at visualization, without clustering, automated validation, or public status tracking (Narciso et al., 2021).

The platforms examined in the speed-and-response literature demonstrate faster resolution but were not evaluated for their handling of geospatial prioritization (Schiff, 2025). The clustering and AI-validation literature is drawn largely from contexts outside community environmental reporting altogether (Xie et al., 2022; Fang et al., 2023), and the transparency commitments documented by Philippine LGUs remain at the planning or pilot stage rather than embedded within an existing citizen-reporting system (Open Government Partnership, 2025). This study addresses that integration gap directly, in line with its statement of the problem. EcoPin A.I. combines a unified map-based reporting interface, a structure that allows lead time over scheduled patrols to be measured, cluster-based hotspot identification, and a public-facing status tracker within a

single, AI-assisted platform, evaluated against the ISO/IEC 25010 quality model to ensure that the integration is not merely feasible but well built.

Cross-Verification and Confidence Scoring for Non-Expert Environmental Data

A separate line of research addresses report credibility through verification methodology rather than incentive design. According to a systematic review of 259 ecological citizen science schemes, verification approaches such as expert cross-checking and community consensus scoring varied considerably across programs but were consistently identified as the primary mechanism for building trust in non-expert-submitted data, more so than participant training or platform design alone (Baker et al., 2021).

According to more recent work applying this approach specifically to environmental monitoring, non-expert contributors in low-resource settings could produce valid water-quality data when supported by structured, simplified verification tools rather than expert oversight alone (Ramesh et al., 2024). For EcoPin A.I., this indicates that credibility can be built into the reporting interface itself, through structured metadata fields, simple confidence prompts, or cross-comparison against nearby reports, rather than through a rewards or reputation-tier system layered on top of it.

Legal and Jurisdictional Limits on Local Waste Enforcement

A separate constraint concerns the authority local government units actually have to act on a citizen report once it is filed, particularly when the reported concern sits on or near private property. According to a study documenting barangay-level implementation of the Ecological Solid Waste Management Act (RA 9003) in Barbaza, Antique, barangay officials generally recognized their obligations under the law but were constrained by irregular collection schedules, non-functional materials recovery facilities, and insufficient funding, producing a persistent gap between the law's mandate and what barangays could enforce in practice (Yazawa et al., 2025).

This evidence indicates that a reporting platform cannot assume that a flagged concern will be acted on simply because it has been made visible, jurisdictional and resource limits at the barangay level shape whether enforcement follows detection at all. For EcoPin A.I., this underscores the importance of routing reports to the correct level of government and being transparent with citizens that a pin's unresolved status may reflect capacity constraints on the LGU side rather than platform failure.

GPS and Vehicle-Tracking Mechanisms for Field Team Coordination

Once an LGU dispatches a field team to verify or act on a cluster of reports, the platform's usefulness depends partly on whether that team's movement can be coordinated and monitored. According to a study detailing the deployment of GPS-based vehicle tracking integrated with real-time location monitoring and secure data handling in a logistics context, continuous location tracking paired with basic security safeguards can support dispatch decisions and accountability for field units (Siddique et al., 2024).

Although this study addresses commercial logistics rather than government field operations specifically, the underlying tracking and dispatch mechanics are directly transferable to an LGU cleanup or inspection crew responding to EcoPin A.I. pins. This literature is included with that caveat, no government-specific GPS/AVL source was found during this search, so the analogy to municipal field teams is drawn from adjacent technical evidence rather than a public-sector study.

Citizen Satisfaction and Perceived Value in E-Government Services

A platform's long-term relevance depends not only on whether it works technically but on whether citizens judge their experience with it favorably enough to keep using it. According to a survey of e-government users in Vietnam, of five service-quality dimensions tested, trustworthiness and fulfillment were the two significant predictors of citizens' perceived value, which in turn

predicted both satisfaction and loyalty toward continued use of e-government services (Pham et al., 2023).

This suggests that a citizen-facing satisfaction or rating feature on EcoPin A.I. would be measuring something with a demonstrated causal path to sustained platform use, rather than serving as a cosmetic add-on. It also implies that improving perceived trustworthiness and follow-through on reports is likely to matter more for retention than expanding the number of features on offer.

Data Privacy and Geolocation Sharing in the Philippine Context

Because EcoPin A.I. depends on capturing a reporter's location to geotag a submission, the platform inherits the same data-privacy questions that have been raised about other location-based Philippine applications. According to a 2024 ACM conference study examining how food delivery platforms operating in the Philippines handle geolocation data under the Data Privacy Act of 2012, consent practices and disclosure of how location data is stored, shared, or retained varied considerably across applications and were not always transparent to end users (Blancaflor et al., 2024).

Applied to EcoPin A.I., this evidence supports building an explicit, plain-language confirmation step before a reporter's precise location is attached to and shared alongside a submitted pin, rather than assuming that location capture is an uncontroversial background function of a mapping platform.

Synthesis

This third cluster of themes addresses the human and institutional layer that surrounds the technical system already described. Report credibility can be built through structured verification and confidence scoring rather than reward-based incentives (Baker et al., 2021; Ramesh et al., 2024), enforcement authority at the barangay level is bounded by resources rather than willingness (Yazawa et al., 2025), field-team coordination has a transferable technical precedent even without a government-specific source (Siddique et al., 2024), citizen satisfaction is driven more by trustworthiness and fulfillment than by feature count (Pham et al., 2023), and location data carries privacy obligations under Philippine law that a reporting platform cannot treat as an afterthought (Blancaflor et al., 2024).

Where the first two clusters established that citizen-contributed, geospatially organized, AI-validated data is technically achievable and institutionally useful, this cluster shows that adoption and legitimacy depend on the platform first earning trust through verifiable, transparent means, respecting jurisdictional limits, and handling personal data responsibly.

Route and Dispatch Optimization for Field Response

Once clusters are identified, an LGU's ability to act on them depends on how efficiently field crews are routed to affected sites, a question directly relevant to EcoPin A.I.'s planned route and task optimization module. According to a study applying metaheuristic vehicle-routing algorithms, including simulated annealing and particle swarm optimization, to the municipal solid waste collection system of

Malolos City, Bulacan, measurable reductions in route length and collection time were demonstrated using a Philippine dataset (Santos et al., 2024). This is a rare source that speaks directly to the national context rather than an international analogy, and it supports treating EcoPin A.I.'s route pathfinder as a validated optimization approach rather than an untested add-on.

Predictive Modeling for Environmental Hotspot Forecasting

EcoPin A.I.'s scope specifies a Random Forest-based hotspot forecasting feature, which requires supporting literature distinct from the descriptive clustering work already discussed (Xie et al., 2022). According to a 2024 ensemble machine-learning study that built a positive-unlabeled random-forest model to predict undiscovered illegal-dumping sites from a limited set of historically confirmed cases, its top-ranked predicted areas captured over two-thirds of subsequently reported new cases (Geng et al., 2024).

This is a close technical analogue to what EcoPin A.I. would need to do which is to forecast likely future hotspots from a necessarily incomplete set of past citizen reports, rather than waiting for a location to accumulate enough reports to be recognized retrospectively.

Equity Gaps in Who Reports: Socioeconomic Patterns in Citizen-Sensor Platforms

A platform's coverage is only as good as who actually uses it, and this raises a caution that EcoPin A.I.'s own scope and limitations section already anticipates. An area may appear problem-free simply because no engaged reporter lives there. According to an analysis of reporting patterns on the UK's FixMyStreet platform, service requests followed a parabolic relationship with neighborhood deprivation, with moderate-deprivation areas reporting the most while the most deprived areas under-reported relative to their actual maintenance needs, and with a small number of "superusers" driving much of the volume in less-deprived neighborhoods (Matthews et al., 2023).

It was concluded that local governments coproducing services through citizen-sensor platforms must actively guard against reports becoming a proxy for where the most engaged residents live rather than where problems are worst. For EcoPin A.I., this means an absence of pins in a given barangay cannot be read by an LGU as an absence of problems without a corresponding measure of reporting density or engagement in that area.

Field-Experimental Evidence on Citizen Reporting for Solid Waste Services Specifically

The most direct empirical test of this study's core premise comes from a field experiment rather than a platform-design study. According to a field experiment in which fifty citizen reporters across one hundred Kampala neighborhoods sent weekly SMS reports on solid waste service delivery to the municipal government over nine months, generating nearly 24,000 reports, citizen reporting did not measurably reduce informal waste accumulation (Buntaine et al., 2021). The qualitative follow-up traced this failure not to citizen apathy but to a mismatch between the inconsistent, unstructured information citizens produced and the decision-making processes municipal managers actually used. This is an important counterweight to the otherwise favorable literature cited earlier in this review (Schiff, 2025; Wolff et al., 2021).

It demonstrates that citizen reporting volume alone does not guarantee institutional improvement unless the data arrives in a form the receiving agency can act on. For EcoPin A.I., this is direct empirical support for why the platform's structured metadata, clustering, and standardized status categories are not incidental design choices but the specific mechanism intended to avoid the failure mode this study documented.

Civic Technology, Institutional Design, and Government Responsiveness

A final question concerns what ultimately determines whether a civic-technology platform changes outcomes rather than simply changing how citizens communicate. According to an examination of four civic-technology platforms across different national contexts, unequal participation does not automatically translate into unequal policy outcomes. Instead, whether a platform's inputs are converted into government responsiveness depends heavily on the institutional design choices built into the platform and the surrounding bureaucratic response process (Mellon et al., 2025). This reframes the significance of EcoPin A.I. beyond a claim about who reports and how often the platform's public status tracker, its routing logic, and its clustering-based prioritization are themselves institutional design choices that will determine whether citizen reports convert into LGU action, independent of how many residents participate.

Synthesis and Research Gap

This fourth cluster addresses what determines whether citizen input converts into institutional action at all. Efficient field response depends on route and dispatch optimization once clusters are prioritized (Santos et al., 2024). Predictive models can anticipate hotspots before they are even reported, extending beyond the retrospective clustering discussed earlier (Geng et al., 2024). Reporting patterns themselves are unevenly distributed across socioeconomic lines, risking a distorted picture of where problems actually concentrate (Matthews et al., 2023). A large-scale field experiment demonstrates that citizen reporting can fail to

improve service delivery entirely when the resulting data does not fit institutional decision-making processes (Buntaine et al., 2021) and, tying these together, institutional design choices rather than participation volume ultimately determine whether inputs become government responsiveness (Mellon et al., 2025). Read as a whole, this cluster functions as a caution as much as a validation. EcoPin A.I.'s structured metadata, clustering logic, dispatch routing, forecasting component, and status-tracking dashboard are not incremental conveniences but the specific design response to documented failure modes in comparable citizen-reporting systems.

Taken together with the two earlier syntheses, this fourth cluster reinforces that the technical and behavioral groundwork reviewed throughout this chapter, participatory GIS reporting, clustering, AI validation, transparency dashboards, verifiable trust mechanisms, jurisdictional constraints, and privacy-conscious data handling, only produces institutional change if it is embedded in a platform whose operational capacity (routing, forecasting) and design choices are deliberately structured to convert citizen input into government response, while remaining alert to equity gaps in who is heard and to the documented risk that reporting volume alone does not guarantee improved outcomes. No single reviewed study integrates all of these components into one system built for Philippine urban environmental reporting specifically. EcoPin A.I.'s contribution is precisely that integration.

Significance of the Study

Local Government Units gain both visibility and speed: a centralized view of where environmental problems are concentrated, and faster detection of emerging issues compared to scheduled patrols. For them, the platform enables both strategic planning (where to focus resources long-term) and tactical response (where to send crews today).

Environmental Organizations receive consolidated, location-based data to support the strategic planning of cleanup initiatives, assessment of pollution severity, and efficient deployment of volunteers.

Citizens gain transparency about environmental conditions in their community and confirmation that reported problems are being tracked and monitored. They can see patterns that affect their neighborhoods and track institutional response.

Future Researchers can use the study as a reference for integrating AI and geospatial analytics into community-based systems, providing a foundational framework for developing similar technologies.

Statement of the Problem

Given the persistent challenges in waste management and the lack of integrated community-based reporting systems, there is a need for a platform that allows for a unified view, enabling citizens and institutions to identify where problems are concentrated and facilitating faster detection and response times. This study addresses how technology, specifically through the integration of crowdsourcing, artificial intelligence, and geospatial analytics, can enhance the identification,

reporting, and transparent tracking of environmental issues within local communities. In line with this, the study is guided by the following general and specific problems:

General Problem:

This study aims to develop and evaluate EcoPin AI. This crowdsourced geospatial platform utilizes artificial intelligence for environmental report validation, geospatial mapping, and rapid institutional detection to support the Pasig City Solid Waste Management Office in improving the transparency, reliability, and efficiency of environmental incident reporting and response.

Specific Problems:

This study seeks to answer the following questions:

1. What artificial intelligence-based validation mechanisms can be integrated into EcoPin A.I. to improve the accuracy, authenticity, and reliability of citizen-submitted environmental reports by identifying duplicate, incomplete, irrelevant, or misleading submissions, and by generating issue-type classification and severity scoring for each validated report?
2. How can geospatial analytics and automated institutional detection be utilized to verify validated environmental reports, determine environmental hotspots, and rapidly notify the appropriate Solid Waste Management Office personnel for timely response?

3. What are the perceptions of the intended users regarding the effectiveness of EcoPin A.I. as a community-based environmental reporting and monitoring platform?
4. What is the level of performance of the artificial intelligence components integrated into EcoPin A.I. in terms of validation accuracy, environmental issue classification, hotspot detection, and institutional notification?
5. What is the level of software quality of EcoPin A.I. based on the applicable ISO/IEC 25010:2023 software quality characteristics?

Scope and Limitations of the Study

This study focuses on design, development, implementation, and usability evaluation of EcoPin A.I., a web- and mobile-enabled crowdsourced geospatial platform that facilitates transparent environmental reporting and rapid institutional detection for the Pasig City Solid Waste Management Office (SWMO). The system consolidates citizen-contributed environmental observations into a unified, map-based interface accessible to both communities and institutions, while enabling rapid geospatial analysis to identify problem hotspots. The platform emphasizes both transparency (public visibility of reported problems and institutional response) and speed (crowdsourced 24/7 monitoring vs. scheduled patrols).

The geographic scope is limited to Pasig City to ensure focused data collection and meaningful evaluation. The evaluation examines both transparency metrics

(data adoption, visibility, user engagement) and speed metrics (detection lag, hotspot identification accuracy, trend analysis effectiveness).

Draggable map, a 2D draggable geospatial map centered on areas surrounding Pamantasan ng Lungsod ng Pasig, which serves as the primary region for environmental data collection and reporting. The coverage is limited to nearby communities to ensure focused data gathering and manageable system evaluation.

Pin creation allows users to create location-based environmental reports by placing pins on the map. Each report includes metadata such as title, description, exact geographic coordinates, status and supporting images or videos that document the environmental condition. Issue type and severity are not fields that the citizen fills in at submission. Both are generated automatically by the AI validation layer immediately after a report is submitted, since non-expert reporters cannot be relied upon to self-classify accurately (for example, a reporter may not recognize that a discarded spray canister falls under special waste).

User roles and access permissions define four categories of platform users, each with distinct capabilities within EcoPin A.I. Citizens can create pin reports, track the status of their own submissions, and receive resolution notifications, but cannot view or modify other users' administrative data. Field Crew members are LGU-affiliated personnel who receive assigned cleanup tasks, update a task's status, including marking it in progress or resolved, and log offline updates while on-site. Officers are SWMO personnel who verify incoming reports, review the

AI-generated severity and issue-type classifications, and assign clusters to Field Crew for action. Admins hold system-wide access, including managing user accounts and role assignments, configuring clustering and prioritization parameters, and viewing platform-wide analytics and the status-tracking dashboard. This role structure ensures that report validation, task assignment, and system configuration are handled only by users with the appropriate level of institutional authority.

Report status and resolution tracking is a feature that allows environmental reports to move through a defined workflow of submitted, verified, assigned, in progress, and resolved. Authorized users such as LGU personnel or verified volunteers may update the status of a report at each stage, after validation, assignment, or cleanup actions. A report is marked in progress once a cleanup crew begins actively working on it, distinguishing active work in the field from a task that has only been assigned. These status changes are reflected in real time on a status-tracking dashboard, giving LGU/SWMO personnel continuous visibility into which tasks are currently in progress, rather than only an end-of-day check. When a report reaches resolved status, the citizen who submitted it is notified and prompted to close the report, at which point they are also asked to rate their satisfaction with the resolution. This feature improves transparency, accountability, and progress tracking within the platform.

Offline report submission and synchronization allows citizens and field-based LGU crews to capture reports and status updates while offline, particularly in areas with inconsistent connectivity. Once network access is restored, locally

stored data automatically synchronizes with the online system, and the real-time map is updated accordingly. This is intended to support field crews who operate on-site and may need to log updates before returning to an area with stable connectivity.

In cases where two or more users update the same report offline and synchronize at or near the same time, the system resolves the conflict by applying the most recent update based on synchronization timestamp.

Daily task monitoring tracks unfinished tasks, pending complaints, and cases not completed by the end of each operational day, giving LGU/SWMO supervisors visibility into daily backlog and enabling timely reassignment of unresolved cases to cleanup crews. This provides an end-of-day operational check separate from real-time hotspot detection, ensuring that cases do not go unaddressed simply because they fell outside the current day's response window.

Location services integration allows the platform to support mobile access and utilize device location services (GPS) to automatically capture the precise location of reported environmental issues, improving accuracy and ease of reporting.

Reporter identity, anonymity, and disclosure consent are handled by retaining reporter identity and address information at the system level for accountability purposes, reducing the risk of fabricated or malicious reports. Reporters may choose whether this identity is disclosed publicly, anonymous reporting is fully accommodated and does not affect a report's validity or processing, and reporter

information is never made publicly visible without consent. Regardless of anonymity settings, the system requires explicit confirmation from the reporter before their location or identifying details are shared or disclosed to any third party, protecting reporters from potential retaliation or conflict, particularly in cases involving reports on or near private property.

AI-based content validation incorporates AI-assisted image and video validation using pre-trained models or APIs to detect inappropriate, irrelevant, or non-environmental content. Beyond flagging a submission as valid or invalid, this same validation stage classifies the underlying issue type, mapping it to the correct RA 9003 category (solid, bulky, or special waste), and assigns a severity score to each report, and the report is separately tagged with an urgency classification that reflects how time-sensitive the issue is (for example, improperly discarded batteries, paint cans, or spray canisters left in a public area would be flagged as high-urgency even if a comparable amount of visible waste elsewhere would not be). Severity captures how serious a condition is, while urgency captures how quickly it needs a response. Both the issue-type classification and the severity score are derived from keywords, location, scale, and risk indicators extracted from the submitted image or video and its accompanying metadata, not from citizen self-tagging or from the number of reports received. Together, these values inform how reports are prioritized within clustering and task assignment. Flagged submissions may be automatically removed or reviewed initially to maintain the integrity of the platform.

Geospatial analysis and hotspot identification applies geospatial analysis techniques such as clustering and density mapping to identify and visualize pollution-prone areas based on user-submitted reports.

Hotspot forecasting predicts when and where possible hotspots will emerge on the map, integrating a machine learning algorithm called "Random Forest" to forecast future environmental hotspots using historical and contextual data.

Cluster-based report resolution improves institutional efficiency by implementing automated clustering of spatially and temporally proximate reports. Reports submitted within 30 minutes and within 50 meters of each other, describing similar issue types, are grouped into clusters visible on the LGU administrative dashboard. LGU personnel can create a single cleanup task linked to all reports in a cluster, which automatically transitions all linked reports to "RESOLVED" status upon task completion. This design reduces administrative overhead and enables batch-processing of environmental issues, aligning system workflows with institutional operational capacity.

Cluster-based complaint summarization generates an automatic summary for each identified cluster, giving personnel a single consolidated overview of community-wide issues rather than requiring manual review of each individual report. Isolated (non-clustered) reports are retained and displayed individually rather than summarized.

Complaint closure and feedback automates the complaint lifecycle from submission to closure, following the workflow: complaint submitted, complaint

acknowledged, agency response logged, resolution marked complete, system sends a reminder prompting the complainant to confirm resolution, complainant rates the resolution using a 5-level emoji-based satisfaction scale (Very Dissatisfied, Dissatisfied, Neutral, Satisfied, Very Satisfied), and complaint is officially closed. The emoji format lowers the input burden on complainants compared to a numerical score, while still producing a simple, consistent metric. This removes the need for personnel to manually follow up with complainants by phone, and gives the LGU/SWMO an aggregated view of resolution satisfaction across complaints.

Route and task optimization module boosts cleanup crews' efficiency by employing a task generator to prioritize the most severe and urgent tasks first and a route pathfinder feature to highlight the best path for the cleaners.

Limitations

The study is limited to environmental concerns directly related to the responsibilities of the Pasig City Solid Waste Management Office. Other environmental issues managed by different government agencies, such as flooding, air pollution, traffic management, water quality monitoring, drainage maintenance, hazardous waste management, wildlife conservation, or disaster response, fall outside its coverage.

The Artificial Intelligence component assists in report validation and environmental issue classification based on the training dataset used in the study. Its predictions support decision-making but do not replace human

verification. Final confirmation, inspection, and action on submitted reports remain under the authority of authorized SWMO personnel.

The geospatial analysis performed by the system is limited to the GPS coordinates available from users' devices. Indoor environments, dense urban structures, or poor satellite reception can reduce positional accuracy, and mapping accuracy may suffer from weak GPS signals, inaccurate device location services, or incorrect location information provided by users.

The AI model recognizes only the environmental issue categories represented in its training dataset. Reports involving unfamiliar, emerging, or mixed environmental conditions may not be classified accurately and require manual verification by authorized SWMO personnel; conditions falling outside the trained categories may need manual review by administrators. Because severity scoring is derived from the same validation output as issue-type classification, it inherits this limitation: a misclassified or ambiguous image can also produce an inaccurate severity score, and both should be treated with the same degree of caution during manual review.

EcoPin A.I. provides offline reporting through delayed synchronization, but several functionalities still depend on internet connectivity. AI-based validation, real-time report transmission, geospatial map updates, hotspot visualization, and administrative notifications all require a stable network connection. Reports submitted offline are temporarily stored on the user's device and uploaded once internet access becomes available, which may delay processing and verification.

Integration with external government information systems, emergency response platforms, national GIS databases, social media platforms, Internet of Things (IoT) sensors, unmanned aerial vehicles (drones), or closed-circuit television (CCTV) surveillance systems is not included in this study, with the exception of the route and task optimization module, which relies on third-party traffic and weather APIs to compute dynamic routing.

Route recommendations are only as accurate and available as these external data providers. Outages, rate limits, or inaccuracies on their end may affect the quality of suggested routes. No other external system integration is included beyond this.

Software quality is evaluated only against the applicable ISO/IEC 25010:2023 characteristics, and AI model performance is assessed using standard machine learning evaluation metrics. Long-term operational effectiveness, predictive environmental forecasting, and continuous model retraining using real-world operational data fall outside the scope of this study.

The system collects only the user information necessary to preserve accountability while limiting exposure to privacy risks. This minimal-data approach reduces the amount of personally identifiable information collected, though it may limit stronger identity verification mechanisms.

Hotspot analysis and the platform's overall effectiveness depend heavily on the volume, quality, and consistency of user-generated reports. Because hotspot identification and trend analysis rely on crowdsourced data, sparse, irregular, inaccurate, or geographically biased submissions can produce incomplete or less

reliable geospatial insights. Areas with little or no user participation may appear free of environmental issues even when unreported incidents are present. Video submissions face the same limitation: AI-based video validation is subject to the same environmental and quality-related constraints affecting photo validation, and typically requires longer processing time given the added complexity of video analysis. Community participation shapes these outcomes directly, as low engagement, inaccurate reporting, or the belief that environmental reporting is solely a government responsibility can reduce both the quantity and quality of submitted reports, weakening the system's ability to detect, monitor, and visualize environmental issues.

The system operates only within the territorial jurisdiction of Pasig City, configured around the operational workflow of the Pasig City Solid Waste Management Office. Deploying it to other cities or municipalities would likely require changes to reporting categories, administrative processes, GIS boundaries, and institutional routing mechanisms.

Real-world outcomes also hinge on the institutional capacity of the LGU/SWMO, particularly its personnel, equipment, and budget. The system enables quicker detection and better visibility of reported issues, but interviews with SWMO personnel revealed that only two cleanup crews are currently available for dispatch, meaning identified hotspots may not always receive prompt action. This study focuses on strengthening detection and visibility; actual remediation depends on institutional resources beyond the system's scope.

Jurisdictional limitations apply as well, particularly for reports involving private property, where field verification requires the property owner's consent before inspection can proceed. If that consent is delayed, withheld, or otherwise difficult to obtain, resolving such reports may take longer than those filed on public land, since cleanup crews and verifying personnel have no legal authority to enter or act on private premises without it. This constraint sits outside the system's control and depends on cooperation among the reporting party, the property owner, and the LGU/SWMO.

The study does not extend to commercial waste collection. EcoPin centers on citizen-generated environmental incident reporting, monitoring, analytics, and hotspot forecasting; commercial waste collection requests, permit processing, waste type eligibility validation, and collection scheduling fall outside its scope and are noted as potential directions for future development.

CHAPTER 2

Methodology

Introduction

This chapter presents the methods, procedures, and tools used to develop and evaluate EcoPin A.I., a crowdsourced geospatial platform designed to improve both the visibility and speed of environmental problem detection in urban areas near Pamantasan ng Lungsod ng Pasig. The purpose of this chapter is to provide a clear and replicable account of how the study was conducted, covering the research design, the architectural framework of the system, the software engineering paradigm and algorithms employed, the population and sampling strategy, the data collection instruments, the data analysis procedures, and the ethical safeguards observed throughout the research.

The study uses a mixed-methods approach because neither quantitative nor qualitative data alone can fully capture how well EcoPin A.I. performs in practice. System logs and platform metrics reveal what the platform does in measurable terms, such as detection speed and hotspot accuracy, while interviews and open-ended survey responses reveal why the system does or does not work for the people who use it. Together, both types of evidence feed back into the iterative development process, making each version of the system more responsive to the actual needs of residents and local government units.

The methodology is organized to directly address the five specific problems identified in Chapter 1: enabling efficient and reliable citizen reporting (Specific Problem 1), identifying environmental patterns and hotspots through AI and geospatial analytics (Specific Problem 2), capturing user perceptions of the platform's effectiveness (Specific Problem 3), evaluating the accuracy of the integrated machine learning algorithms (Specific Problem 4), and assessing software quality against ISO/IEC 25010:2023 (Specific Problem 5). Each methodological choice, from the selection of Agile Scrum as the development framework to the use of geospatial clustering and AI-based image validation, is tied to these research objectives. This alignment ensures that the evaluation of EcoPin A.I. goes beyond measuring system usage and instead demonstrates whether the platform produces faster, more equitable, and more transparent environmental action.

Research Design

This study uses a Developmental Research Design to design, create, implement and evaluate EcoPin A.I. Developmental research is appropriate because the primary goal of the study is to produce a functional software solution that solves an identified problem while systematically evaluating its quality and effectiveness. The software was developed using the Agile Software Development Life Cycle. Agile emphasizes continuous stakeholder collaboration, adaptive planning, and frequent delivery of functional software. This approach enabled the researchers to progressively develop system modules, conduct regular consultations with the

project adviser and the client, evaluate completed features and incorporate feedback throughout the development process. Such flexibility was very important because the system requirements evolved during the study, specifically in the implementation of AI-image based validation, geospatial analysis, report verification and user interface improvements.

Other development methodologies were also considered. The Waterfall Model follows a sequential process in which each development phase must be completed before the next begins. While suitable for projects with fixed and well-defined requirements, it offers limited flexibility when modifications arise during development. Since EcoPin A.I. required continuous refinement based on stakeholder feedback and emerging functional requirements, the Waterfall Model was not chosen.

The Rapid Application Development (RAD) Model emphasizes rapid prototyping and accelerated software development through extensive user involvement. Although RAD is effective for quickly developing user interfaces and prototypes, it is generally more acceptable for projects with simpler technical requirements. EcoPin A.I. incorporates AI-powered image validation, geospatial analytics, database management, and environmental data visualization, which require iterative testing, validation, and integration beyond rapid prototyping alone.

Considering the evolving project requirements, continuous stakeholder involvement, and the need for incremental development and testing, the Agile Method was selected as the most appropriate development methodology.

Its iterative nature enabled the continuous improvement of the system, validating implemented features and ensuring that the final software met both user requirements and the applicable ISO/IEC 25010:2023 software quality characteristics.

Software Engineering Paradigm

The development of software systems is commonly guided by the Software Development Life Cycle (SDLC), a framework that provides a structured process for planning, designing, developing, testing, deploying, and maintaining software applications. Following a well-defined development methodology helps ensure that the resulting system meets user requirements while maintaining software quality throughout its development.

For this study, the researchers adopted the **Agile Software Development Life Cycle (Agile SDLC)** as the software engineering paradigm in developing **EcoPin A.I.** Unlike traditional linear development models, Agile promotes an iterative and incremental approach where system features are continuously developed, evaluated, and refined based on stakeholder feedback. This methodology was appropriate for the study because the system requirements evolved throughout the development process due to consultations with the project adviser and the client, the Solid Waste Management Office (SWMO) of Pasig City. Through Agile, the researchers were able to validate completed features, incorporate suggested improvements, and address identified issues in every development iteration.

The development of EcoPin A.I. followed the Agile SDLC through the following phases:

The planning phase served as the foundation of the project. During this stage, the researchers identified the objectives, scope, limitations, and feasibility of the proposed system. To better understand the existing waste management process, the researchers reviewed publicly available information about the SWMO, examined current waste management practices, and explored existing technologies and related studies. These activities helped establish the project's initial direction and identify the problems that EcoPin A.I. aimed to address.

Requirements analysis phase involved gathering and analyzing the functional and non-functional requirements of the system. Interviews and consultations were conducted with the Head of the Solid Waste Management Office to understand the department's existing workflow, operational challenges, and desired system features. The collected requirements were then translated into system specifications that guided the succeeding stages of development. Continuous consultations with the project adviser and the client also allowed the researchers to refine and update the requirements whenever necessary.

System design phase built on the approved requirements, as the researchers designed the overall structure of EcoPin A.I. This included planning the system architecture, database structure, user interface, and system workflows. Decisions regarding the implementation of features, artificial intelligence components, algorithms, and development technologies were also made during this stage. The

design served as the blueprint that guided the implementation of the system while ensuring that both technical requirements and user needs were addressed.

The implementation phase covered the development of the system based on the finalized design. This involved coding the different modules of EcoPin A.I., integrating the database and AI functionalities, and developing both the administrator and citizen interfaces. Development was carried out incrementally, allowing completed features to be tested and improved before proceeding to subsequent functionalities. This iterative approach enabled the researchers to accommodate changes requested by the client and adviser throughout the development process.

Testing phase covered the various testing activities the completed modules underwent to verify that the system functioned according to the specified requirements. Unit testing and integration testing were performed during development, while consultations and demonstrations with the client served as opportunities for user validation. Feedback gathered from these sessions was used to correct defects, improve usability, and refine system features before deployment. This iterative testing process contributed to the reliability and overall quality of the developed system.

Deployment phase began after successful testing and validation, when EcoPin A.I. was prepared for deployment. This phase involved configuring the system for operational use and ensuring that it was ready to support the Solid Waste Management Office and the citizens who would utilize its services. The

deployment process also ensured that the implemented features operated correctly within the intended environment.

The maintenance phase, the final phase, focuses on the continuous improvement of the system after deployment. As organizational needs and technologies evolve, EcoPin A.I. may require corrective maintenance, feature enhancements, and performance improvements. Since the system is intended for the Solid Waste Management Office, future maintenance may be performed by the agency or its designated technical personnel to ensure that the system remains functional, secure, and responsive to future operational requirements.

Architectural Framework

This framework will outline the structure of the EcoPin A.I. System, including the interaction with residents, local government units (LGUs), the interface, back-end services, and its database.

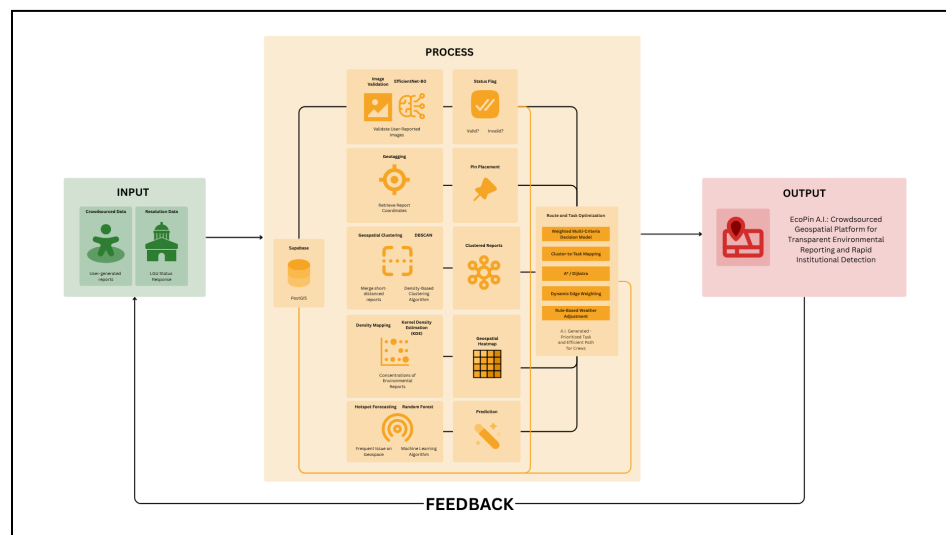


Figure 1: Architectural Framework

As shown in Figure 1, the system is organized into an Input layer, a Process layer, and an Output layer connected by a feedback loop. The **Input layer** captures two categories of information that enter the system from opposite ends of the reporting cycle.

Crowdsourced Data consists of user-generated reports submitted through the resident-facing interface. Each report includes a photo or video of the environmental concern, a geotagged pin marking its location, and a short description. This is the raw signal the system is built to act on. These data are unverified, unstructured, and variable in quality, which is precisely why it must pass through validation before it can be trusted for decision-making.

Resolution Data consists of status updates submitted by LGU personnel once a reported issue, or a cluster of related issues— has been acted upon. This data re-enters the system as confirmation that a task has been completed, and it is what ultimately powers the feedback loop back to residents.

Both data streams converge at the same entry point into the Process layer, ensuring that citizen input and institutional response are handled through a single, consistent pipeline rather than two disconnected systems.

The **Process layer** is where EcoPin does the bulk of its analytical work. It corresponds to the Application Layer (workflow logic, task orchestration) and the Data Layer (Supabase with PostGIS), and it houses the AI Layer, which performs

the four learning-based tasks that give EcoPin its predictive and organizational capability. It is best understood as five sequential stages:

Image validation (AI layer: EfficientNet-B0) begins with every submitted photo being passed through EfficientNet-B0, a convolutional neural network trained for image classification. Functionally, the network extracts visual features from the image (texture, object shapes, color distribution) and outputs a probability score indicating whether the photo genuinely depicts a relevant environmental issue such as uncollected waste and littered plastic. This produces a Status Flag of Valid or Invalid. In the same pass, and only for reports flagged Valid, the stage also outputs an issue-type classification mapped to the corresponding RA 9003 category (solid, bulky, or special waste) and a severity score, so that classification and severity are produced together with the validity determination rather than as separate downstream steps. Reports flagged invalid are filtered out before they can distort downstream clustering or hotspot analysis. This is the system's first line of defense against noise and misuse.

Geotagging and pin placement takes place once a report passes validation, when its coordinates are retrieved and rendered as a pin on the interactive map. This step is largely a data-handling operation rather than an inference task: it takes the latitude/longitude captured at submission and anchors the report spatially, making it queryable for every subsequent geospatial process.

Geospatial clustering (AI layer: DBSCAN) groups validated reports using Density-Based Spatial Clustering of Applications with Noise (DBSCAN). Unlike clustering methods that require a fixed number of clusters to be specified in

advance, DBSCAN groups points based on spatial density, that means— reports that fall within a defined proximity (50 meters) and time window (30 minutes) of one another, and that describe similar issue types, are merged into a single cluster. Reports that don't meet this density threshold are left as standalone points ("noise"), rather than forced into an ill-fitting group. The output is a set of Clustered Reports, which allows the system to recognize that ten residents photographing the same overflowing garbage bin is one problem, not ten.

Density mapping (AI layer: Kernel Density Estimation) runs in parallel, applying Kernel Density Estimation (KDE) across the full set of validated reports to produce a Geospatial Heatmap. KDE works by placing a smooth probability distribution ("kernel") over each report's location and summing these distributions across the map, producing a continuous surface that shows where environmental complaints are most concentrated. This differs from clustering in intent: DBSCAN groups discrete reports for task creation, while KDE visualizes overall spatial concentration for pattern recognition and dashboard reporting.

Hotspot forecasting (AI layer: Random Forest) feeds historical report data that includes report frequency, location, issue type, and seasonality into a Random Forest model, an ensemble learning algorithm that builds multiple decision trees on different subsets of the data and aggregates their outputs to produce more stable and reliable predictions than a single decision tree. The model identifies areas that are more likely to become future environmental hotspots, enabling Local Government Units (LGUs) to shift from a purely reactive approach (responding only after reports are filed) toward a more preventive and

risk-reduction-oriented strategy by allocating resources to areas with elevated risk.

However, the generated forecasts are intended only as decision-support recommendations and should not be interpreted as definitive predictions. The identified hotspot areas require manual validation by authorized personnel, who must consider current field conditions and other relevant factors before taking action. As emphasized by the stakeholders, the forecasting component is designed to assist planning and prioritization rather than replace human judgment or official decision-making.

Route and Task Optimization. The outputs of clustering, density mapping, and forecasting converge in this final stage, where clustered reports are translated into concrete, schedulable work for LGU crews:

A **Weighted Multi-Criteria Decision Model** scores clusters against factors such as severity, report volume, and public health risk, producing a prioritized task order.

Cluster-to-Task Mapping converts each prioritized cluster into a single administrative task linked to all of its constituent reports.

A* / Dijkstra pathfinding algorithms compute the most efficient travel route between task locations for field crews.

Dynamic Edge Weighting adjusts the "cost" of travel routes in real time based on road conditions or traffic.

Rule-Based Weather Adjustment applies predefined rules to reprioritize or reroute tasks when weather conditions (e.g., heavy rain) affect accessibility or urgency.

The result is an AI-generated, prioritized task list with an efficient route already computed for the crew assigned to execute it— accounting for SWMO's 2 sole teams when capacity is exceeded.

Underlying all of these stages is Supabase (PostGIS), the Data Layer component that stores and indexes every report's geographic coordinates, images, and status, and provides the spatial querying capability that clustering, density mapping, and routing all depend on.

Output layer is where the Process layer's outputs converge into EcoPin's core deliverable: a crowdsourced geospatial platform that gives residents transparent visibility into environmental reporting while giving LGUs a rapid, AI-supported system for institutional detection and response. In practice, this output manifests as the resident-facing hotspot map and notifications, and the LGU-facing dashboard of prioritized, validated, route-optimized tasks.

Feedback loop activates when an LGU marks a task complete, as that Resolution Data flows back into the Input layer, automatically updating the status of every report linked to the resolved cluster. This closes the loop: residents see their reports move to "Resolved," and the system retains a historical record that continues to feed the Random Forest forecasting model, allowing hotspot predictions to improve over time as more resolution outcomes accumulate.

Algorithm Discussion

EfficientNet-B0 (Image & Video Validation)

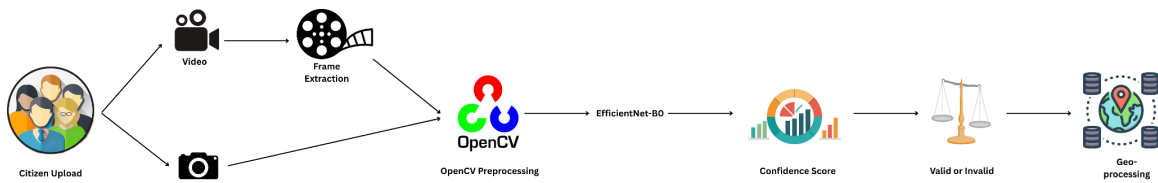


Figure 2: Overall EfficientNet-B0 Validation Pipeline

Figure 2 illustrates how every submitted image or video frame moves through preprocessing, feature extraction, and classification before a Valid/Invalid status is returned. EfficientNet-B0 is a convolutional neural network (CNN) used in EcoPin to validate both uploaded images and videos before they enter the geospatial analysis pipeline. While images are classified directly, videos undergo an additional preprocessing stage where representative frames are extracted and individually validated. This enables EcoPin to support richer evidence while preventing irrelevant or misleading multimedia submissions from affecting downstream geospatial analysis.

When a resident submits a report, the accompanying media is first preprocessed using OpenCV. Frames on videos are extracted and each and every one of them will be preprocessed just like images. The preprocessing stage standardizes the image by resizing it to 224×224 pixels, converting it into the required RGB format, and normalizing pixel values to match the input requirements of

EfficientNet-B0. Standardizing every image ensures that differences in camera resolution, aspect ratio, or device type do not affect the model's predictions.

Video Frame Extraction. Unlike static images, videos contain hundreds or even thousands of frames, many of which are visually similar. Processing every frame would significantly increase computational cost while providing little additional information. To improve efficiency, EcoPin first extracts representative frames from each uploaded video using OpenCV. Frames are sampled at fixed intervals (for example, one frame every second or every n frames), reducing redundancy while preserving the important visual content throughout the video. Each extracted frame is treated as an independent image and undergoes the same preprocessing steps applied to image submissions, including resizing to 224×224 pixels, RGB conversion, and pixel normalization before being forwarded to EfficientNet-B0.

After preprocessing, each image, or each extracted video frame, is forwarded to EfficientNet-B0 for feature extraction. Since every frame is processed independently, the same CNN architecture can be reused for both image and video validation without requiring a specialized video recognition model.

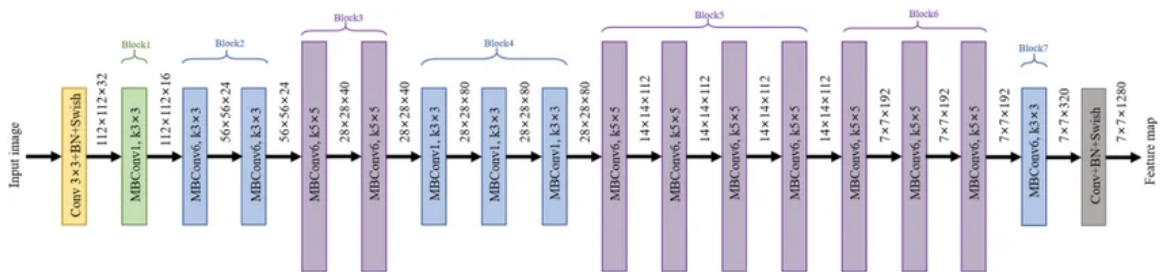


Figure 3: EfficientNet-B0 Architecture

This pipeline is powered by the EfficientNet-B0 architecture, depicted in Figure 3, which consists of three major components: the Stem, Body, and Head.

The **Stem** serves as the initial feature extraction stage. It begins with a 3×3 convolution containing 32 filters and a stride of 2.

The 3×3 kernel allows the network to examine small neighborhoods within the image and capture simple visual patterns such as edges, corners, color transitions, and object boundaries. Using a stride of 2 reduces the spatial resolution by moving the filter two pixels at a time, decreasing computational cost while preserving the most important visual information.

The extracted feature maps are then passed through Batch Normalization, which stabilizes training by normalizing intermediate activations, followed by the ReLU6 activation function, which introduces non-linearity while remaining efficient for lightweight mobile architectures.

The **Body** performs the majority of the visual analysis through a sequence of Mobile Inverted Bottleneck Convolution (MBConv) blocks.

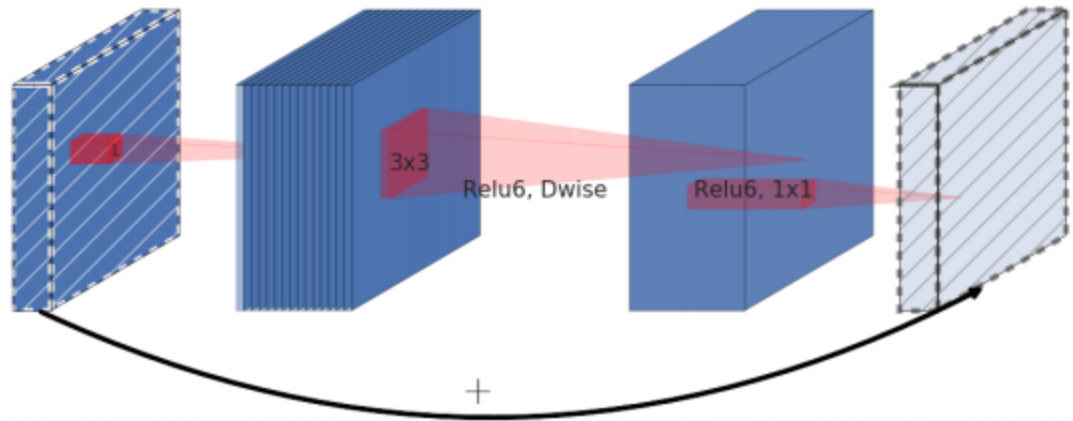


Figure 4: Structure of an inverted residual block (MBconv)

Figure 4 details the internal structure of each MBConv block, which follows an expand-depthwise-project pattern with a residual connection. Unlike traditional convolutional layers, MBConv blocks employ depthwise separable convolutions, significantly reducing the number of parameters and computational operations while maintaining high classification accuracy.

Each MBConv block consists of four stages:

Expansion Layer (1×1 Convolution). The input channels are expanded into a higher-dimensional feature space, allowing the network to learn more complex visual representations.

Depthwise Convolution. Instead of applying every filter across every channel, one filter is applied independently to each channel.

This greatly reduces computational complexity while preserving spatial information.

Squeeze-and-Excitation (SE) Block. EcoPin benefits from the SE block because it allows EfficientNet-B0 to emphasize the most informative feature channels while suppressing less relevant ones. For example, when validating environmental reports, the model can automatically assign greater importance to visual characteristics associated with overflowing garbage, illegal dumping, construction debris, or bulky waste items left curbside while reducing attention to irrelevant background objects such as buildings, roads, or pedestrians.

Projection Layer (1×1 Convolution). The expanded feature maps are compressed back into a compact representation before being passed to the next MBConv block.

This narrow → wide → narrow structure enables EfficientNet-B0 to achieve high classification accuracy while remaining computationally efficient enough for near real-time validation.

After passing through multiple MBConv blocks, the extracted feature maps enter the **Head**. The Head consists of a final 1×1 convolution, Global Average Pooling, and a fully connected classification layer.

Global Average Pooling summarizes each feature map into a single representative value, greatly reducing the number of parameters compared to traditional fully connected layers.

The final dense layer applies the Softmax activation function, producing a probability distribution across all predefined environmental classes.

Example Output Class	RA 9003 Category	Severity Output
Garbage Accumulation	Solid Waste	Low – High, scored from image and metadata
Illegal Dumping	Solid Waste	Low – High, scored from image and metadata
Discarded Batteries/Paint	Special Waste	Typically High, given associated risk indicators
Discarded Furniture	Bulky Waste	Low – High, scored from image and metadata
Non-environmental image	Not applicable (Invalid report)	Not applicable
Blurry/low-quality image	Not applicable (Invalid report)	Not applicable

Table 1: EcoPin's Output Classes

The class with the highest probability becomes the predicted label. If the prediction corresponds to a recognized environmental issue and its

confidence score exceeds the configured threshold, the report is marked Valid and proceeds to the geospatial processing pipeline.

Otherwise, the report is flagged Invalid, preventing irrelevant images from affecting hotspot detection and environmental analytics.

Frame-Level Classification. For video submissions, EfficientNet-B0 generates an independent prediction for every extracted frame.

Each frame produces:

Predicted Class

Confidence Score

Probability Distribution

Frame	Prediction	Confidence
1	Garbage	96%
2	Garbage	94%
3	Road	42%
4	Garbage	95%

Table 2: Sample Frame Confidence Classification

Because individual video frames may contain motion blur, camera shake, or temporary occlusions, EcoPin performs confidence voting across all validated frames. After EfficientNet-B0 classifies every extracted frame, the system

aggregates the predictions using majority voting together with average confidence scores.

Class	Frames
Garbage	18
Bulky Waste	2
Invalid	1

Table 3: Sample Aggregation of Predictions using Confidence Voting

Since the majority of extracted frames depict garbage accumulation, the video is classified as a valid garbage report. If most extracted frames are classified as non-environmental content or if the average confidence falls below the configured threshold, the entire video is marked as invalid. This voting mechanism reduces the influence of noisy or low-quality frames and produces a more reliable multimedia validation result.

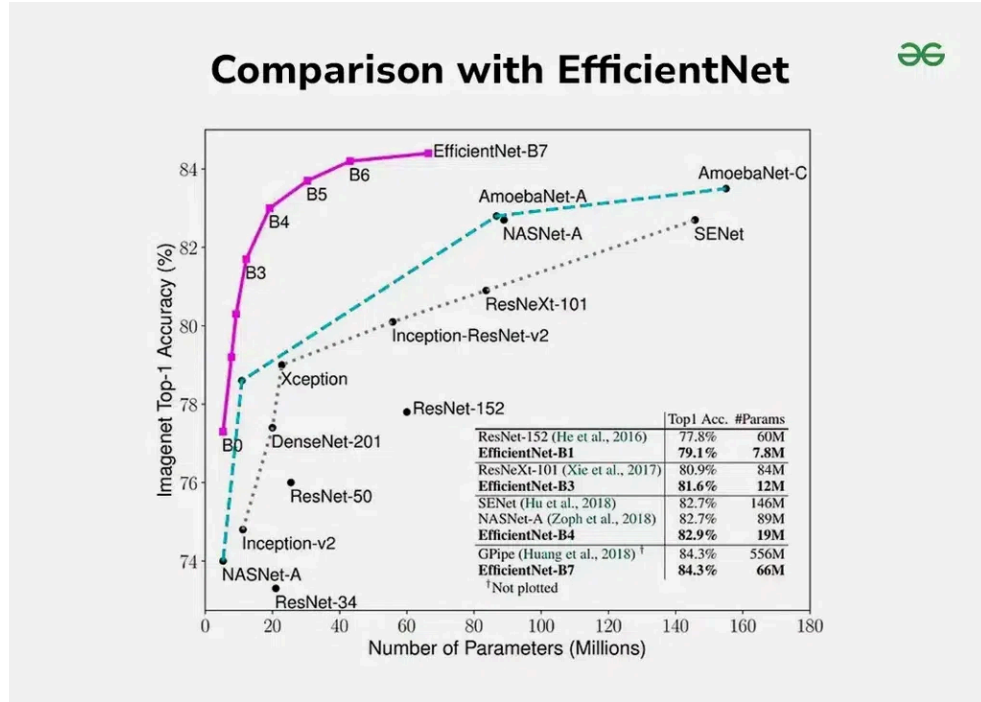


Figure 5: Official EfficientNet Performance Evaluation and Comparison

EfficientNet-B0 was selected because it provides an effective balance between classification accuracy and computational efficiency. Compared with larger CNN architectures such as ResNet-50 or DenseNet, EfficientNet-B0 requires substantially fewer parameters while maintaining competitive accuracy through its compound scaling strategy and MBConv architecture. This makes it suitable for EcoPin, where uploaded images must be validated quickly without imposing excessive computational overhead on the server. Furthermore, EfficientNet-B0 supports transfer learning, allowing the model to be fine-tuned using environmental images collected for EcoPin instead of requiring a large dataset to be trained from scratch.

To make this possible, the tools and libraries used are: **TensorFlow**, which is the main framework used to build and train the neural network, **Keras** wherein it works on top of TensorFlow and makes it easier to design and fine-tune the model and finally, **OpenCV**, for image preprocessing tasks like resizing and format conversion.

It is worth clarifying that EfficientNet-B0, as described, is a pure image classifier operating over pixel data; it does not itself read keywords, location, or other non-visual fields. The severity score produced during validation is therefore not generated by the CNN alone: it combines the CNN's visual classification output with non-visual metadata already captured at submission (geotag, timestamp, and description text) rather than implying that the vision model directly parses text or coordinates.

DBSCAN (Geospatial Clustering)

DBSCAN groups spatial points based on local density rather than requiring the number of clusters to be specified in advance. In EcoPin, the algorithm is applied to unresolved environmental reports through PostgreSQL/PostGIS's built-in ST_ClusterDBSCAN window function, which clusters report locations directly inside the database.

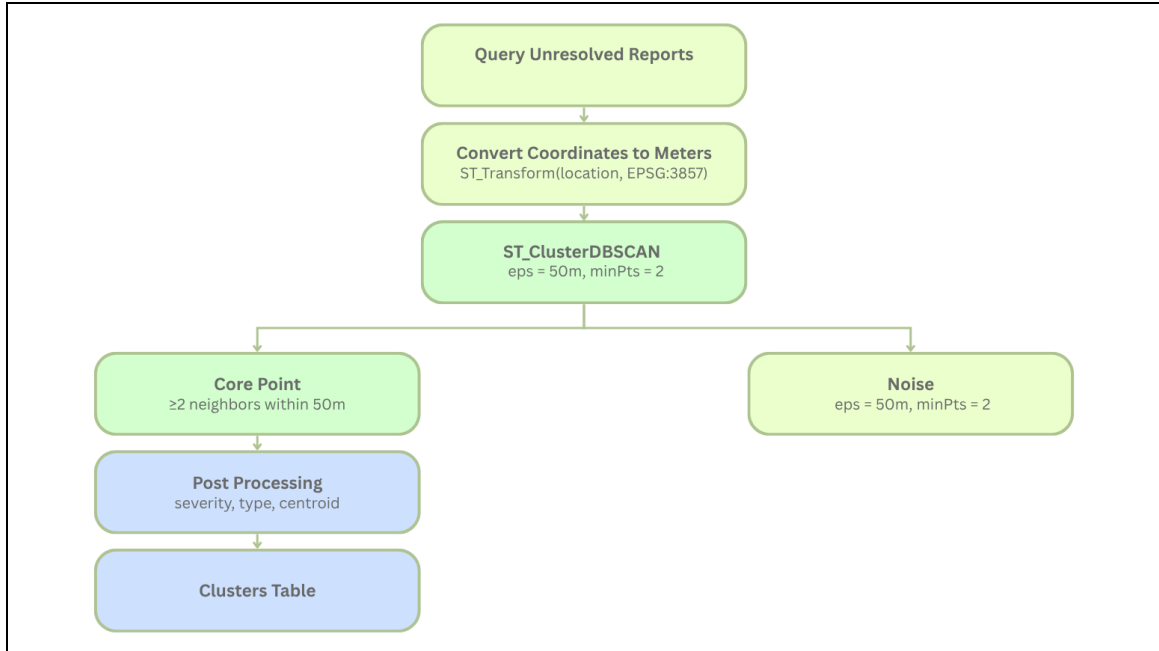


Figure 6: DBSCAN clustering flow, from unresolved report to the clusters table.

As shown in Figure 6, the clustering process begins with a query that retrieves all unresolved reports from the database. Each report's location is then converted from geographic coordinates into a planar coordinate system measured in meters, which is required for ST_ClusterDBSCAN to interpret the eps parameter correctly. The transformed points are passed into ST_ClusterDBSCAN, which evaluates each report against its neighbors: a report with at least minPts neighbors within eps meters is classified as a core point and included in a cluster, while a report that does not meet this threshold is classified as noise and assigned a cluster_id of NULL. Reports that form a cluster proceed to post-processing, where the system determines severity and dominant issue type before the cluster is stored in the clusters table; noise reports are excluded from this stage entirely.

EcoPin uses $\epsilon = 50$ meters and $\text{minPts} = 2$. Since `ST_ClusterDBSCAN` measures distance in whatever unit the input geometry is in, report coordinates first go through `ST_Transform` to convert them from geographic coordinates (EPSG:4326, in degrees) into Web Mercator (EPSG:3857, in meters). Without that step, $\text{eps} := 50$ would be read as 50 degrees rather than 50 meters.

`ST_ClusterDBSCAN` is a built-in PostGIS function backed by the GEOS library. It takes the transformed point geometries along with `eps` and `minpoints`, runs the neighbor search and core-point classification internally in C, and returns a `cluster_id` for each row, with NULL marking noise.

As an example, consider four hypothetical unresolved reports that are geographically close to one another, with coordinates measured in meters:

Report	x(m)	y(m)
A	0	0
B	30	20
C	45	-15
D	300	300

Table 4: Coordinates of Sample Unresolved Reports

Using the standard Euclidean distance formula, the distance between reports A and B is approximately 36.06 m, between A and C is approximately 47.43 m, and between B and C is approximately 38.08 m. Since all three distances are below the 50 m threshold, the reports are considered neighbors. In contrast, report D is

approximately 424 m away from report A, placing it well outside the specified neighborhood radius.

$$dist(p, q) = \sqrt{(x_p - x_q)^2 + (y_p - y_q)^2}$$

Because report A has two neighboring reports (B and C) within 50 m, it satisfies the minimum point requirement ($minPts = 2$) and is therefore classified as a core point. Report D, having no neighboring reports within the specified radius, is treated as noise and is assigned a `cluster_id` value of NULL.

After the clustering process is completed, `clustering.service.js` processes the results by grouping reports according to their `cluster_id` while excluding reports classified as noise. For each cluster, the system determines the dominant environmental issue by identifying the most frequently occurring `issue_type` (here, `issue_type` refers to the AI-assigned classification produced by the EfficientNet-B0 validation stage for each report, not a field entered by the citizen). It then assigns a cluster-level severity by aggregating the AI-assigned per-report severity scores already produced for each member report during validation, taking the highest per-report severity within the cluster as the cluster's severity level. The number of reports in the cluster is used only as a secondary, tie-breaking factor, and is no longer the primary basis for determining severity.

The service also invokes `upsert_cluster_for_reports`, which uses PostGIS to calculate the geometric centroid of all report locations within each cluster. For a collection of points, the centroid corresponds to the average of their coordinates.

The computed centroid, together with a fixed 50 m radius, is then stored in the clusters table. In the example above, reports A, B, and C form a cluster containing three reports, resulting in a Medium severity classification.

The selection of $\text{minPts} = 2$ is lower than the values commonly used in general-purpose spatial clustering, where a minimum of four or five points is often recommended. This decision reflects a deliberate design tradeoff for a crowdsourced environmental reporting platform. In EcoPin, two independent residents reporting the same location within a 50 m radius are considered sufficient evidence to indicate a potential environmental hotspot. This approach enables earlier detection of time-sensitive issues, such as waste accumulation and illegal dumping, instead of delaying identification until more reports are submitted. While a lower minPts increases the likelihood of false positives during the clustering stage, the system mitigates this risk by relying on the assigned severity level to determine whether a hotspot should be escalated to LGU personnel for further action.

Kernel Density Estimation / KDE (Density Mapping)

EcoPin's heatmap layer (EcoPinMap.js) is implemented using leaflet.heat, which visualizes report density through a grid-based approximation of Kernel Density Estimation (KDE) rather than a full continuous KDE. The rendering process consists of two stages. First, HeatLayer aggregates report points into a spatial grid. Second, simpleheat renders the grid onto an HTML5 canvas as overlapping radial gradients, which are then mapped to a predefined color scale.

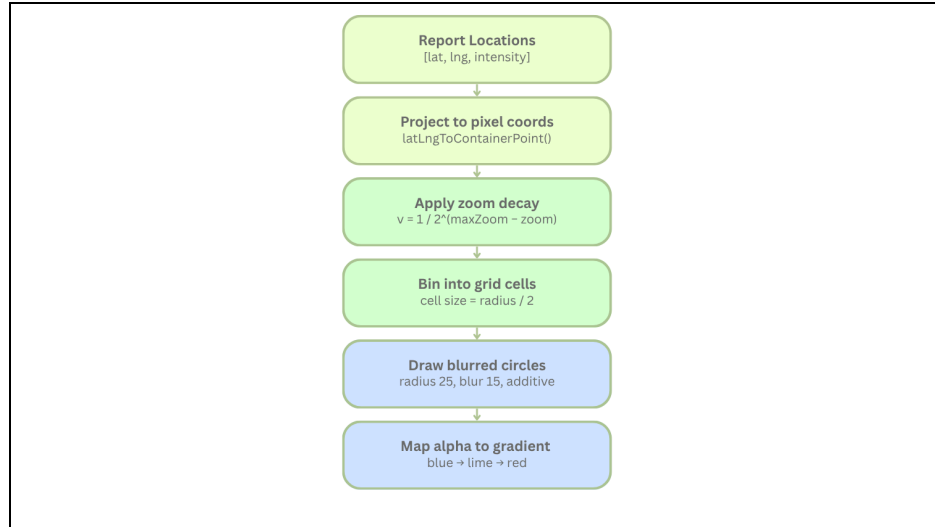


Figure 7: Heatmap rendering flow, from raw report location to colorized pixel.

As shown in Figure 7, each report begins as a raw location with an associated intensity value, which is projected into pixel coordinates on the current map view. That projected point is then scaled by a zoom-dependent decay factor, so its influence on the map shrinks as the user zooms out. The scaled point is added to a spatial grid, where it's merged with any other points falling in the same cell rather than treated individually. Once all points are binned, each occupied grid cell is rendered as a blurred circle, and overlapping circles blend additively to build up density. Finally, the accumulated intensity at every pixel is mapped through a fixed color gradient to produce the blue-to-red heatmap.

Each unresolved report is initially converted into a [latitude, longitude, intensity] tuple, where the intensity defaults to 1.0. Before any density calculations are performed, each point is projected from geographic coordinates into pixel coordinates relative to the current map view using `latLngToContainerPoint()`. Any

point that falls outside the visible viewport, including a buffer equal to the configured radius, is excluded from further processing to avoid unnecessary rendering computations.

To ensure that report density remains visually consistent regardless of the map's zoom level, each point's intensity is multiplied by a zoom-dependent decay factor before being added to the spatial grid:

$$v = \frac{1}{2^{\max(0, \min(\text{maxZoom} - \text{currentZoom}, 12))}}$$

With maxZoom set to 17, points contribute at full intensity when the map is displayed at the maximum zoom level because maxZoom – currentZoom = 0, resulting in (v = 1). As the user zooms out, the value of (v) decreases exponentially. For example, at currentZoom = 15, the difference becomes 2, yielding (v = $\frac{1}{2^2}$ = 0.25). This scaling prevents reports displayed at lower zoom levels from visually overwhelming the map, which would otherwise occur if raw, unscaled intensities were used.

The projected points are then grouped into a spatial grid whose cell size is equal to radius / 2, corresponding to 12.5 pixels with the configured radius of 25 pixels. Instead of evaluating a kernel function at every pixel, as performed in a true KDE implementation, each grid cell maintains a running weighted centroid. The first point assigned to a cell is stored directly. Each subsequent point updates the centroid using a weighted moving average while also increasing the cell's accumulated intensity according to the following equations:

$$cell.x = \frac{cell.x \cdot oldK + px \cdot k}{oldK + k}$$

$$cell.y = \frac{cell.y \cdot oldK + py \cdot k}{oldK + k}$$

$$cell.intensity = oldK + k$$

Here, k represents the weighted intensity contributed by a report after applying the zoom-dependent scaling factor ($k = \text{intensity} \times v$), while oldK represents the accumulated intensity already stored within the grid cell.

As an example, consider two reports displayed at zoom level 15, where ($v = 0.25$). Suppose the reports project to pixel coordinates (100, 102) and (105, 98), with both falling inside the same 12.5-pixel grid cell. The first report is stored directly with:

Step	Calculation	Result
First report weight	$k_1 = 1.0 \times 0.25$	0.25
Initial cell state	First report stored directly	(100, 102, 0.25)
Second report weight	$k_2 = 1.0 \times 0.25$	0.25
Updated x-coordinate	$((100 \times 0.25) + (105 \times 0.25)) \div (0.25 + 0.25)$	$51.25 \div 0.5 = \mathbf{102.5}$
Updated y-coordinate	$((102 \times 0.25) + (98 \times 0.25)) \div (0.25 + 0.25)$	$50 \div 0.5 = \mathbf{100}$
Accumulated intensity	$0.25 + 0.25$	0.5

Table 5: Sample Calculation of Grid Cell Centroid and Accumulated Intensity

After all reports have been assigned to their respective grid cells, simpleheat renders one blurred circular gradient at the centroid of each occupied cell using a 25-pixel radius and a 15-pixel blur. The blur softens the edge of each circle, producing smooth transitions between neighboring density regions instead of sharp boundaries.

These circles are rendered using additive blending, allowing overlapping influence from nearby cells to accumulate rather than overwrite one another. The opacity of each circle is computed as:

$$opacity = \frac{\min(cell.intensity, max)}{max}$$

where max = 1.0. For the example above:

$$opacity = \frac{\min(0.5, 1.0)}{1} = 0.5$$

After all circles have been drawn, the canvas maps each pixel's alpha value (0–255) using the following fixed color gradient:

{ 0.4: 'blue', 0.65: 'lime', 1.0: 'red'}

In the previous example, an opacity value of 0.5 falls between the 0.4 and 0.65 gradient thresholds and is therefore rendered as a color interpolated between blue and lime, indicating a low-to-moderate density region rather than a hotspot. Only cells whose accumulated intensity exceeds the 0.65 threshold are displayed using lime-to-red colors, making dense concentrations of reports visually distinguishable from isolated incidents.

This grid-and-blur technique approximates Kernel Density Estimation rather than implementing a true Gaussian KDE. A conventional Gaussian KDE evaluates a kernel function across every pixel in the map, making it computationally expensive when thousands of points must be redrawn continuously during panning and zooming. By first grouping reports into a coarse spatial grid and then representing each occupied cell with a blurred radial gradient, leaflet.heat achieves real-time rendering while still producing an effective visualization of report density. The primary tradeoff is spatial resolution. Reports that fall within the same 12.5-pixel grid cell are merged into a single blended representation instead of being displayed individually. Although this reduces positional precision, it is well suited for identifying city-scale environmental hotspots, where overall spatial trends are more important than building-level accuracy.

The heatmap is rendered as a toggleable overlay (`showHeatmap` state in `EcoPinMap.js`, line 201) rather than serving as the default map view because it is intended to complement the clusters produced by DBSCAN.

While DBSCAN identifies where a cleanup or response team should be deployed, the heatmap highlights where environmental problems are currently most concentrated, providing users with an overall view of report density across the city.

Random Forest (Hotspot Forecasting)

Random Forest is a machine learning algorithm that works by building multiple decision trees and combining their outputs to produce a more reliable prediction. In EcoPin, Random Forest will be used to identify areas that have a higher likelihood of becoming future environmental hotspots based on historical patterns, providing LGUs with a decision-support tool for preventive planning rather than simply displaying locations where issues have already occurred.

This capability is valuable because relying solely on reports after environmental problems have accumulated can delay response efforts. By identifying areas with elevated risk, LGUs will be able to prioritize inspections, allocate resources more efficiently, and implement preventive measures before issues worsen. However, the generated forecasts are intended to support planning and risk reduction only. They should not be interpreted as definitive predictions and must be manually validated by authorized personnel, who should consider current field conditions, recent developments, and other relevant information before making operational decisions.

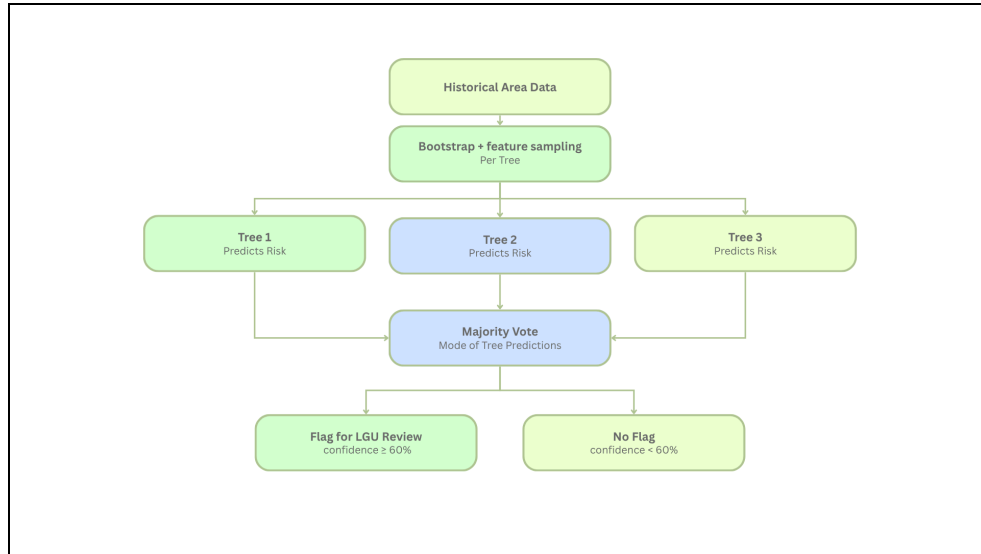


Figure 8: Random Forest hotspot forecasting flow, from historical area data to a flagged or unflagged classification

As shown in Figure 8, historical data for each area will first be drawn into bootstrap samples, with each tree also restricted to a random subset of features. Every tree will independently arrive at its own risk prediction based on the patterns it has learned. These individual predictions will then be combined through a majority vote to produce a single forecast along with a confidence score, and only areas whose confidence score meets the configured threshold will be flagged for LGU review.

The model will be trained using historical information associated with each area, including report frequency, issue severity, cleanup history, weather conditions such as rainfall (used as a predictor of waste accumulation and dumping likelihood, since heavy rain can disperse litter and debris into public areas, rather than as a predictor of flood occurrence), traffic patterns, time of day, nearby

establishments, and population density. Rather than relying on a single decision-making path, Random Forest will construct multiple decision trees, each trained on a bootstrap sample and each considering only a random subset of the available features at every split, rather than all of them. This two-part randomness (bagging the data and subsampling the features) is what keeps the trees from all learning the same patterns, so their errors tend to cancel out rather than compound.

Each individual tree will grow by repeatedly splitting the data on the feature and threshold that best separates hotspot cases from non-hotspot cases. The quality of a candidate split is commonly measured using Gini impurity:

$$Gini = 1 - \sum p_i^2$$

where p_i is the proportion of samples in a node belonging to class i (in EcoPin's case, "hotspot" or "not a hotspot"). A node containing only one class has a Gini impurity of 0 (perfectly pure), while a 50/50 mix has the maximum impurity of 0.5.

As an illustrative example, consider a tree evaluating whether to split a set of 60 historical area-records on the condition "rainfall > 50mm," used here as a proxy for how much litter and debris heavy rain disperses into public areas rather than as an indicator of flooding. Of the 40 records with rainfall at or below 50mm, suppose 30 were not waste-accumulation hotspots and 10 were; of the 20 records with rainfall above 50mm, suppose 4 were not hotspots and 16 were.

The Gini impurity of each resulting group would be:

Rainfall $\leq$ 50mm:

$$p(\text{not hotspot}) = \frac{30}{40} = 0.75$$

$$p(\text{hotspot}) = \frac{10}{40} = 0.25$$

$$Gini = 1 - (0.75^2 + 0.25^2) = 1 - 0.625 = 0.375$$

Rainfall $>$ 50mm:

$$p(\text{not hotspot}) = \frac{4}{20} = 0.2$$

$$p(\text{hotspot}) = \frac{16}{20} = 0.8$$

$$Gini = 1 - (0.2^2 + 0.8^2) = 1 - 0.68 = 0.32$$

The tree would combine these into a single weighted impurity for the split, weighted by how many records fall into each side:

$$\text{Weighted Gini} = \frac{40}{60} \times 0.375 + \frac{20}{60} \times 0.32 = 0.25 + 0.1067 \approx 0.357$$

The tree would compare this value against the impurity of other candidate splits (different features or thresholds) and select whichever split produces the lowest weighted Gini impurity, since a lower value indicates a split that more effectively separates hotspot from non-hotspot records.

Once every tree in the forest has been grown this way, a prediction for a new area will be made by passing its feature values through all the trees and combining their individual votes. The final classification is the majority vote:

$$\hat{y} = \text{mode}\{T_1(x), T_2(x), \dots, T_n(x)\}$$

where $T_i(x)$ is the prediction of the i -th tree for input x , and n is the total number of trees in the forest. As an illustrative example, suppose the forest contains 100 trees, and for a given area, 70 of them predict "high risk" while 30 predict "low risk." The majority vote would classify the area as high risk, with the proportion of agreeing trees ($70/100 = 70\%$) used as a confidence score for that prediction. EcoPin will flag an area for LGU review only when this confidence score meets or exceeds a set threshold so a narrow or uncertain split among the trees (say, 52 votes to 48) will not trigger an automatic escalation.

In order to build this feature, the team will be using several Python libraries. Scikit-learn will be used to build, train, and evaluate the Random Forest model. Pandas will be used to organize, preprocess, and clean the historical dataset prior to training, while NumPy will support the numerical computations required during data processing and model execution.

This design reflects a deliberate tradeoff between responsiveness and reliability. A single decision tree trained on the same data could produce a hotspot prediction faster, but it would also be far more sensitive to noise or unusual cases in the training set. By training many trees on different random samples of the data and features, and requiring agreement across a majority of them before a prediction is trusted, the forest is expected to be more resistant to any one anomalous pattern, at the cost of requiring more historical data and computation than a single tree would.

Route and Task Optimization

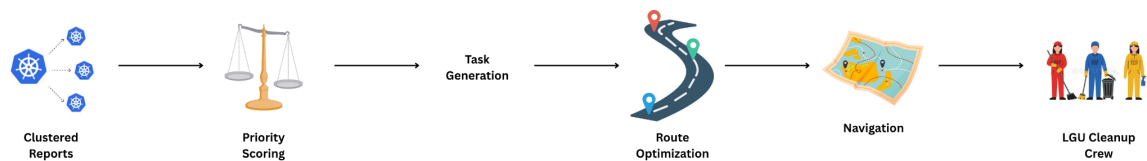


Figure 9: Route and Task Optimization Pipeline

After validation, reports are processed through prioritization, task mapping, routing, and dynamic adjustment before reaching the assigned cleanup crew, as shown in Figure 9. Unlike the previous AI components of EcoPin, Route and Task Optimization is not implemented using a single machine learning algorithm. Instead, it combines several optimization and graph-search algorithms that work sequentially to transform validated environmental reports into practical cleanup tasks for the Solid Waste Management Office (SWMO).

The objective of this module is not only to determine which environmental issues should be addressed first, but also how limited cleanup crews can travel between

these locations as efficiently as possible. Since interviews with the SWMO indicated that only two cleanup teams are available, efficient prioritization and routing are essential for maximizing institutional response capacity.

The optimization process consists of five major stages:

Weighted Multi-Criteria Decision Model (Task Prioritization). Once DBSCAN has grouped nearby reports into environmental clusters, EcoPin must determine which cluster should receive attention first. Treating every report equally could result in crews responding to minor litter complaints while more urgent situations, such as hazardous waste, continue to worsen.

To address this problem, EcoPin employs a Weighted Multi-Criteria Decision Model (WMCDM). Rather than relying on a single factor, the model evaluates several characteristics of each cluster simultaneously and combines them into one overall priority score.

Typical evaluation criteria include:

Severity level assigned by AI validation (aggregated at the cluster level)

Number of citizen reports within the cluster

Urgency classification

Waiting time since submission

Public health or environmental risk

Current weather conditions

Each criterion is assigned a numerical weight based on its relative importance.

The overall priority score is calculated as:

$$Priority = \sum_{i=1}^n (Weight_i \times Score_i)$$

where each criterion contributes proportionally to the final ranking.

For example, consider two environmental clusters:

Cluster	Severity	Reports	Waiting Time	Priority
Uncollected Bulky Waste	High	7	2 hours	Very High
Illegal Dumping	Medium	3	1 day	High

Table 6: Sample Environmental Clusters

Note: The Severity values shown above are not derived from report count. They are aggregated at the cluster level from the AI-assigned per-report severity scores produced during EfficientNet-B0 validation, consistent with the revised DBSCAN severity logic described earlier.

Although the illegal dumping report has existed longer, the uncollected bulky waste cluster receives a higher priority because it presents greater immediate risk to public safety. This ranking produces a prioritized task queue that determines the order in which cleanup operations should be scheduled.

Cluster-to-Task Mapping. Once clusters have been ranked, EcoPin converts each cluster into a single operational task. Without clustering, ten residents reporting the same illegal dumping site would generate ten independent cleanup requests. This would unnecessarily duplicate administrative work. Instead, EcoPin maps every report belonging to the same cluster into one cleanup task.

Each generated task contains:

Cluster centroid

Dominant issue type

Number of linked reports

Severity

Priority score

Assigned cleanup crew

Current task status

This current task status field, including the in-progress state, is what feeds the real-time status-tracking dashboard described earlier, so LGU/SWMO personnel can see which tasks are actively being worked on at any moment rather than only at the end of the day. When the task is completed, all reports linked to that cluster automatically receive updated status notifications. This many-to-one mapping significantly reduces workload while ensuring every resident receives feedback regarding the resolution of their report.

Vehicle Routing Problem (VRP) is one of the most extensively studied optimization problems in operations research. After determining which tasks should be completed, EcoPin must determine how the available cleanup crews should travel between them.

The Vehicle Routing Problem was originally introduced as the truck dispatching problem, in which the goal was to determine optimal routes for a fleet of vehicles delivering goods from a central depot, and it has since grown into one of the most widely researched problems in operations research and logistics optimization (Tan & Yeh, 2021). The objective is to determine the most efficient routes for a limited number of vehicles while minimizing travel distance, travel time, or operational cost.

In EcoPin:

Vehicles represent SWMO cleanup crews.

Customers represent prioritized environmental tasks.

Depot represents the SWMO office.

Travel cost represents estimated driving distance or travel time.

Because only two cleanup crews are available, solving the routing problem helps maximize daily coverage while reducing unnecessary travel.

The optimization objective is to minimize:

$$\sum TravelDistance$$

subject to available crews, working hours, vehicle capacity and task priorities. Rather than manually assigning routes, EcoPin automatically generates efficient task schedules using Google OR-Tools.

In **A* Search Algorithm (Shortest Path Computation)** after determining the sequence of assigned tasks, EcoPin computes the shortest travel path between successive locations using the A* search algorithm.

A* is a heuristic-based best-first search algorithm that remains one of the most widely used pathfinding methods in robotics, GPS navigation systems, video games, and autonomous vehicle path planning, and it continues to serve as a foundational technique in modern mobile robot

and vehicle navigation research (Hanxi et al., 2021). Unlike Dijkstra's Algorithm, which explores all possible directions equally, A* incorporates a heuristic estimate of the remaining distance to the destination, allowing it to focus its search toward the target.

The evaluation function is:

$$f(n) = g(n) + h(n)$$

where:

$g(n)$ is the travel cost already incurred,

$h(n)$ is the estimated remaining distance,

$f(n)$ is the estimated total travel cost.

For EcoPin:

$g(n)$ represents the road distance already traveled.

$h(n)$ is estimated using the straight-line distance between the current road segment and the cleanup location.

By combining these two values, A* typically reaches the destination faster than uninformed search methods while still guaranteeing the shortest path when an admissible heuristic is used. For example, if two roads lead toward the same cleanup task, A* naturally favors the road that appears to move closer to the destination while still considering accumulated travel

cost. This enables cleanup crews to navigate efficiently between multiple environmental tasks.

Dynamic Edge Weighting. Real-world road conditions constantly change due to traffic congestion, flooding, road closures, or construction. If routing decisions relied solely on physical distance, the shortest route might actually require more travel time than a longer but uncongested alternative. To address this, EcoPin applies Dynamic Edge Weighting.

In graph theory, every road is represented as an edge connecting two intersections (nodes). Instead of assigning a fixed distance to every edge, EcoPin continuously adjusts edge costs using real-time traffic and environmental information.

Road	Normal Cost	Heavy Traffic	Flooded
Road A	5	12	25
Road B	8	8	8

Table 7: Sample Traffic and Environmental Information

Although Road A is physically shorter, heavy flooding increases its travel cost. Consequently, A* selects Road B because its total travel time becomes lower. This allows route optimization to reflect actual road conditions rather than theoretical distances.

Rule-Based Weather Adjustment. Some environmental hazards become significantly more urgent under specific weather conditions. Unlike

machine learning algorithms that infer patterns from historical data, EcoPin employs a rule-based expert system for immediate weather-dependent adjustments. The system follows predefined IF THEN rules.

Examples include:

IF rainfall exceeds the configured threshold AND the cluster is located near a canal or waterway THEN increase task priority, since heavy rain can wash uncollected garbage into waterways.

IF heavy rainfall is detected AND the destination road is impassable THEN recompute the route.

IF severe weather warning is active THEN prioritize solid-waste clusters near canals and waterways, since typhoon debris tends to increase bulky-waste volume in these areas.

Rule-based systems are particularly suitable because these operational policies are already established by domain experts and do not require model training. This allows EcoPin to rapidly adapt routing decisions based on current environmental conditions while maintaining transparent and explainable decision-making.

Within EcoPin, Route and Task Optimization serves as the final decision-support component before institutional response.

Validated reports are first grouped into clusters by DBSCAN. The Weighted Multi-Criteria Decision Model prioritizes these clusters according to their severity and urgency. Cluster-to-Task Mapping then consolidates related reports into operational cleanup tasks. Google OR-Tools solves the Vehicle Routing Problem to assign tasks to the available cleanup crews, while the A* algorithm computes efficient travel routes between task locations. Dynamic Edge Weighting continuously updates road costs based on traffic and environmental conditions, and Rule-Based Weather Adjustment modifies priorities whenever hazardous weather is detected.

By integrating these optimization techniques, EcoPin enables the SWMO to allocate limited personnel more efficiently, reduce unnecessary travel, prioritize high-risk environmental incidents, and respond more effectively to citizen-generated reports. Rather than replacing human decision-making, the module functions as an intelligent decision-support system that assists LGU personnel in planning daily cleanup operations using real-time geospatial information.

In order to build this entire feature, the team will be using a couple of tools so that implementation would be much smoother. To solve the Vehicle Routing Problem (VRP) - **Google OR-Tools** which is an operations research library is going to be used. For providing real-time routing and navigation data, the researchers could use either: **Google Maps Directions API, OpenRouteService or OSRM**. Finally, for supplying live traffic and weather information, publicly available **Traffic APIs and Weather APIs** are going to be utilized for dynamic route adjustments.

Data Flow Diagram

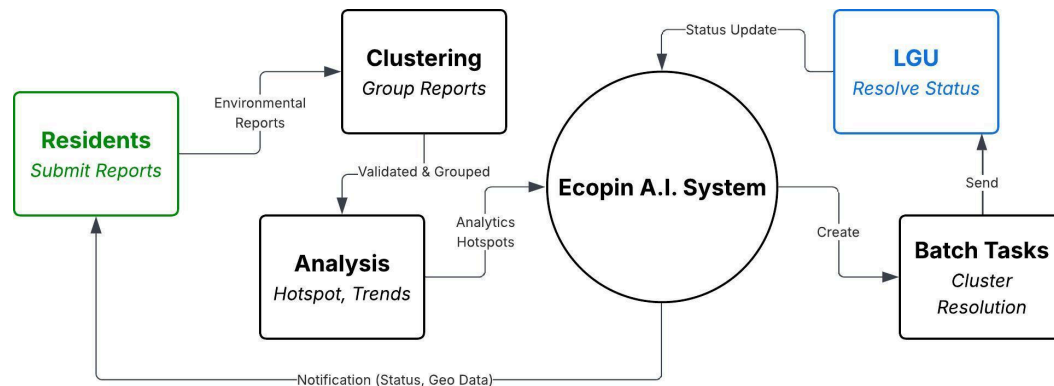


Figure 10: DFD Level 0

The Data Flow Diagram (DFD) Level 0, presented in Figure 10, shows the interaction between residents, the EcoPin A.I. System, and local government units (LGUs). Residents submit environmental reports and account data to the system, which processes the information and generates hotspot map data for user awareness.

The system also forwards reports and analytics to LGUs for monitoring and action. In return, LGUs provide status updates, which are sent back to residents as notifications and report status updates. This flow displays the system's role in enabling efficient reporting and both transparency and fast detection.

Population and Sampling

A robust evaluation of EcoPin AI requires input from all key stakeholders. Therefore, our study engages three distinct groups:

1. LGU officials in the **Solid Waste Management Office**, who are critical for institutional action.
2. Residents of Pasig City, who are the primary end-users.
3. IT experts, who provide technical validation of the system.

The study will use tailored sampling strategies for each group:

For the specialized knowledge of **LGU officials (SWMO)**, we will use Purposive Sampling to identify and invite 5-8 key individuals whose roles directly relate to waste management and community initiatives. Of the 5-8 LGU officials recruited through Purposive Sampling, a subset consisting of the SWMO Head and 2-3 senior personnel directly involved in dispatch and field operations will additionally participate in semi-structured interviews. This qualitative component captures operational context, such as day-to-day dispatch decision-making and institutional constraints, that a Likert-scale instrument alone cannot surface.

For **residents**, a minimum of 30 participants will be recruited using Purposive and Snowball Sampling. This target is grounded in Creswell and Creswell's (2023) minimum threshold for mixed-methods research. Participants will be initially recruited through community centers and online groups, with early participants asked to refer others.

For technical validation, the study will use Purposive Sampling to identify and invite a minimum of 2 **IT experts**, selected based on relevant professional experience (minimum of 3 years) in software/mobile application development,

systems architecture, or AI implementation. These experts will assess the technical soundness of EcoPin AI.

Data Collection Methods

This study uses a survey questionnaire as its primary method of data collection. Instead of designing original evaluation questions, the study adopts the standardized quality characteristics defined under ISO/IEC 25010:2023, the international standard for software product quality evaluation. Since these questions are already established, tested, and widely used across software research, the study does not need to undergo a separate questionnaire validation process with another statistician or research expert because the credibility of the instrument is already backed by the standard itself.

ISO/IEC 25010:2023 defines several quality characteristics that can be used to evaluate a software system, but not all of them apply to every type of system. For EcoPin A.I., only the characteristics that are relevant to how the platform actually functions and how users interact with it were selected. Based on the nature of the system, the following characteristics were identified as applicable:

Functional Suitability - whether the platform's features (report submission, pin placement, clustering, hotspot mapping, task routing) actually meet the needs of its users.

Performance Efficiency - how well the system performs in terms of response time and resource usage, especially when handling image validation, clustering, and route computation.

Interaction Capability (Usability) - how easy and intuitive the platform is to use for both residents and LGU personnel, covering aspects like learnability and user satisfaction.

Reliability - whether the system consistently performs its functions without failure, particularly for report submission and status tracking.

Security - how well the platform protects user data and prevents unauthorized access, which is relevant given that reports contain user-submitted location and image data.

Maintainability - how easily the system can be updated or modified, which is more relevant to the developers/administrators evaluating the platform's backend structure.

Characteristics such as Compatibility, Flexibility, and Safety were excluded, since they relate more to cross-platform deployment, hardware/software environment variations, and physical/operational hazard contexts that fall outside the scope of what this study aims to evaluate.

The survey questionnaire, built around these selected characteristics, will be distributed to the platform's target respondents which are the residents who have used the reporting feature, LGU personnels who interact with the administrative dashboard and IT experts that can meaningfully judge the technical aspects of the system. Each item will be rated using a Likert scale, allowing the responses

to be measured and analyzed quantitatively. Because the items are sourced directly from ISO/IEC 25010:2023, they already reflect internationally recognized standards for measuring software quality, which strengthens the reliability of the results without requiring additional external validation.

Semi-Structured Interviews

To complement the survey questionnaire, semi-structured interviews will be conducted with the SWMO Head and senior personnel identified in the Population and Sampling section. The interview guide will center on five areas, current dispatch and resource-allocation practices, perceived usefulness of the clustering and route-optimization outputs, operational constraints not captured by the survey items (crew availability, jurisdictional limits), how the proposed Admin, Officer, Field Crew, and Citizen role permissions align with actual day-to-day institutional practice, and suggestions for refining the platform's institutional workflow. Interviews will be audio-recorded with consent and transcribed for analysis.

System Log and Platform Metrics Extraction

In addition to self-reported perceptions, the study collects objective performance data directly from EcoPin A.I.'s system logs. This includes timestamps marking the interval between a report's submission and its inclusion in a validated cluster (detection lag), the interval between cluster creation and task resolution (response time), and the count of reports per cluster over the data-collection period. These log-derived metrics provide a quantitative, non-self-reported basis

for evaluating the platform's speed and hotspot-identification claims, independent of survey responses.

Data Analysis Procedures

The survey questionnaire based on ISO/IEC 25010:2023 is the primary data source for this study, so the analysis procedure focuses mainly on quantitative methods appropriate for interpreting Likert-scale responses; the semi-structured interviews and system log data described above are analyzed separately, as detailed later in this section.

Once responses are collected, they will first be tallied and organized per quality characteristic (e.g., Functional Suitability, Performance Efficiency, Interaction Capability, Reliability, Security, Maintainability). Each item under these characteristics is rated using a Likert scale, so the analysis will rely primarily on descriptive statistics, specifically, computing the frequency, percentage, and mean (weighted mean) of responses for each item and for each overall characteristic.

This allows the study to summarize, in numerical form, how respondents generally rated the platform across each of the selected ISO quality characteristics.

To interpret these computed means, a standard Likert scale interpretation range will be used (e.g., a mean falling between 4.20–5.00 interpreted as "Strongly Agree" or "Excellent," down to 1.00–1.79 interpreted as "Strongly Disagree" or "Poor"). This gives each numerical result a clear, understandable descriptive

label rather than leaving it as just a raw number, making it easier to determine whether the platform performed well or poorly under each characteristic.

If the survey includes any open-ended items, for instance, asking respondents to briefly explain a low rating or suggest an improvement then responses to those specific items will be reviewed and grouped into simple, general categories to identify common feedback. This is not a full qualitative analysis, but a light-touch grouping meant only to support and give context to the quantitative results, since qualitative analysis is not a primary requirement of this study.

The goal of this analysis is to determine, in measurable terms, how the EcoPin platform performs across the selected ISO/IEC 25010:2023 quality characteristics, based directly on user and LGU personnel responses.

Analysis of Interview Data

Interview transcripts will be analyzed using thematic analysis. Transcripts will be read in full, coded for recurring concepts related to dispatch practice, platform usefulness, and institutional constraints, and grouped into themes. These themes will be used to contextualize and explain patterns observed in the survey results, particularly for characteristics such as Functional Suitability and Reliability, where SWMO personnel are the primary respondents.

Analysis of System Log Data

Log-derived metrics (detection lag, response time, reports per cluster) will be summarized using descriptive statistics, including mean, median, and range, to quantify the platform's actual detection and response speed during the evaluation period. Where relevant, these figures will be compared against baseline estimates of the SWMO's prior scheduled-patrol response times, if such records are made available, to support the study's core claim of improved detection speed.

Ethical Considerations

This study recognizes that it involves not just human respondents, but also location-based data submitted through the platform itself which carries its own privacy sensitivities separate from the usual concerns in survey research. Because of this, ethical handling of data was treated as a central design concern throughout the study, not just a procedural requirement.

No Collection of Personal Data Without Consent. No personal information, whether from survey respondents or platform users will be collected without the individual's knowledge and consent. Survey participants will be given a clear explanation of what data is being collected and why, and participation will only proceed once they've given their consent. Similarly, platform users submitting environmental reports will be informed, through the platform's terms of use, of what information is collected (e.g., location, images) and how it will be used.

Interview Consent and Recording. Participants in the semi-structured interviews will be informed of the interview's purpose, that it will be audio-recorded, and how the recording and transcript will be stored and used, prior to giving consent. Participants may decline to have the session recorded, request that specific responses be excluded from the transcript, or withdraw at any point without penalty. Recordings and transcripts will be stored securely and accessible only to the researchers directly involved, and any direct quotations used in the study will be reviewed to ensure the SWMO Head or named personnel cannot be identified without their explicit permission.

Confidentiality and Privacy. The study recognizes that report location data, while necessary for the system to function, can be sensitive. This addresses a specific concern raised by SWMO regarding user location privacy during project consultations. It is mentioned that some users may be hesitant to have their reports tied back to their identity or exact whereabouts, particularly if the reported issue is near their home or workplace. To address this, users will have the option to submit reports anonymously, meaning their account identity will not be publicly tied to the reports they file. Any data used in this study's analysis or published as part of the research will be presented in aggregate or de-identified form, ensuring that no individual respondent or user can be identified from the results.

Ethical Handling of Respondent Information. All information gathered through surveys will be used strictly for the purposes of this study and will be stored securely with restricted, role-based access, limited only to the researchers

directly involved. Data will not be shared with third parties or used beyond the scope explained to participants during the consent process.

Future Use of Crowdsourced Data for Model Improvement. Since the platform's AI components (e.g., image validation, clustering, forecasting) may benefit from continued training on real-world data, the study acknowledges the possibility that crowdsourced reports could be used in the future to improve these models. To ensure this is done ethically, this will only apply to users who explicitly opt in and consent to their data being used for this purpose. Even then, any data used for model improvement will be anonymized and stripped of identifying information beforehand, meaning it cannot be traced back to the individual who submitted it. This approach follows the principles of the Data Privacy Act of 2012 (RA 10173), which permits the use of personal data for legitimate purposes such as research and system development, provided that consent is obtained and adequate safeguards are in place to protect the data subject's identity.

Privacy and Anonymity. User identities will be anonymized in all published findings. The platform is designed with privacy-by-design principles, giving users control and visibility over how their location data is used, and the option to remain anonymous when submitting reports.

Data Security. All collected data, whether from surveys or the platform itself, will be protected through secure storage, encryption, and restricted access controls, ensuring information is not exposed to unauthorized parties.

Beneficence. The study is designed so that its contribution to more transparent and responsive environmental governance outweighs the minimal risks posed to participants, particularly given the safeguards in place for privacy and data handling.

Summary

This chapter detailed the developmental research design and methodological framework used to build and evaluate the EcoPin A.I. platform, encompassing the Agile software engineering paradigm, the integration of specific artificial intelligence and geospatial algorithms, and the mixed-methods approach for data collection and analysis. Building upon these established procedures and ethical guidelines, Chapter 3 presents the evaluation results derived from the ISO/IEC 25010:2023 surveys, system logs, and semi-structured interviews, followed by a comprehensive discussion of these findings to determine the platform's overall effectiveness in streamlining environmental reporting and institutional response for Pasig City.

CHAPTER 3

Results and Discussions

Introduction

This chapter presents the results of the development and evaluation of EcoPin A.I., a crowdsourced geospatial platform for transparent environmental reporting and rapid institutional detection for the Pasig City Solid Waste Management Office. The primary purpose of this study is to develop and evaluate a platform that utilizes artificial intelligence for environmental report validation, geospatial mapping, and rapid institutional detection to improve the transparency, reliability, and efficiency of incident reporting and response.

This chapter contains the findings obtained during system development, the extraction of system logs and platform metrics, the performance evaluation of the integrated machine learning models, and the software quality evaluations conducted with Pasig City residents, Solid Waste Management Office personnel, and IT experts. The findings are presented according to the specific problems and objectives of the study. The chapter begins by detailing the capabilities of the AI-based validation mechanisms and the geospatial analytics utilized for automated hotspot detection. This is followed by the results regarding the intended users' perceptions of the platform's effectiveness as a community-based monitoring tool, and the performance level of the AI components in terms of accuracy, classification, and notification.

The chapter presents the system's software quality evaluation based on the applicable ISO/IEC 25010:2023 characteristics. These findings objectively report the collected data without interpretation, serving as the basis for determining whether EcoPin A.I. met the identified requirements and achieved the general and specific objectives of the study.

Project Capabilities and Testing Process Details

Project Capabilities

This section presents the capabilities and major features of the developed EcoPin A.I. crowdsourced geospatial platform. The system was developed based on the requirements gathered during consultations with the Pasig City Solid Waste Management Office (SWMO) and was designed to support environmental report validation, geospatial mapping, hotspot forecasting, and route optimization through artificial intelligence and machine learning technologies.

Project Testing Capabilities

CHAPTER 4

Conclusions and Recommendations

Introduction

Conclusion

Recommendations

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APPENDICES

OPERATIONAL DEFINITION OF TERMS

EVALUATION SHEETS

PROJECT TIMELINE OR GANTT CHART

WORK ASSIGNMENTS

CERTIFICATE OF APPRECIATION

SOURCE CODE

PLAGIARISM CHECK RESULTS

AI GENERATED CONTENT CHECK RESULTS

EXTENDED ABSTRACT

CURRICULUM VITAE

